Software Requirements Specification (SRS)

VidyaConnect — Sri Lanka School Communication Platform

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1. Introduction

1.1 Purpose of the Document

This Software Requirements Specification (SRS) document defines the complete functional and non-functional requirements for VidyaConnect, a mobile-first school communication platform designed for Sri Lankan schools. It provides a detailed and authoritative description of the system's intended behaviour, constraints, and interfaces.

The purpose of this document is to serve as the primary reference for all stakeholders involved in the design, development, testing, and deployment of the VidyaConnect platform. It establishes a shared understanding of what the system must do and how it must perform, reducing ambiguity throughout the software development lifecycle.

This document is intended to be used:

- By the development team as the basis for system design and implementation
- By the testing team as the basis for writing test cases and acceptance criteria
- By project supervisors and mentors for review, feedback, and approval
- By school stakeholders to validate that the proposed system meets their real-world communication needs

This SRS covers the full feature scope of VidyaConnect, including core MVP features, extended-scope features, and optional/future-scope features. The classification of each feature into these categories is detailed in Section 8.

1.2 Product Name

Sri Lankan schools currently rely on a fragmented mix of WhatsApp groups, printed circulars, paper consent forms, verbal messages relayed through students, and occasional phone calls to manage communication between schools, teachers, parents, and students. These tools are ill-suited to the scale and structured nature of school administration. Important notices get buried in group chat activity, paper forms are lost in transit, teachers receive no confirmation that messages have been read, and parents with multiple children must simultaneously monitor multiple disconnected channels.

VidyaConnect (meaning "Knowledge Connect" derived from the Sanskrit word “Vidya”, meaning knowledge or education, used across both Sinhala and Tamil communities in Sri Lanka) is a structured, role-aware, mobile-first communication platform that replaces these fragmented channels with a single trusted source of information for all school stakeholders.

The platform provides:

- A structured Communication Hub replacing WhatsApp groups and printed circulars
- A Digital Assignment and Homework Tracker with automated reminders
- A Digital Consent and Forms Management system replacing paper-based workflows
- A Smart Notification System that eliminates information overload
- A Unified School Calendar visible across all roles
- An Insights Dashboard tailored to each user role
- Bulk Data Management tools for efficient school administration
- Student Progress and Academic Records management
- An AI Assistant, built using a Retrieval-Augmented Generation (RAG) architecture, that allows parents, teachers, and students to ask natural language questions and receive accurate, data grounded answers drawn directly from the school's own records

VidyaConnect is delivered as a React Native mobile application (Android-first, with iOS support) for parents, students, and teachers, and a Next.js web portal for school administrators and teachers. The system is designed specifically for the device constraints, connectivity conditions, and digital literacy levels of the Sri Lankan school context.

About the AI Assistant (RAG based)

A key differentiating feature of VidyaConnect is its built in AI Assistant, accessible as a conversational chat interface within the parent, teacher, and student applications. Rather than relying solely on a general purpose language model, the AI Assistant is built on a Retrieval Augmented Generation (RAG) architecture. This means that whenever a user asks a question, the system first retrieves the relevant, authorised records from VidyaConnect's own database, such as a specific child's assignments, attendance, consent form status, or upcoming events, and provides this real data to the underlying AI model before it generates a response.

This approach ensures that the AI Assistant's answers are accurate, current, and strictly limited to data the requesting user is authorised to access, in accordance with the platform's existing role based access control rules (see Section 5, User Roles and Permissions). It prevents the AI from generating incorrect or fabricated information ("hallucinations") and ensures every response is grounded in verified school records rather than general knowledge.

1.3 Project Scope

1.3.1 In Scope

VidyaConnect's scope spans three categories of features, as detailed in Section 8 (Feature Scope Classification):

Core MVP Features include the Communication Hub, Assignment and Task Tracking, Smart Notification System, Digital Consent and Admin Forms, School Calendar and Event Management, Insights Dashboard, and Bulk Data Management.

Extended Scope Features include Attendance Management, School Notice Board, Student Progress and Academic Records, Parent-Teacher Meeting Scheduler, School Community and Engagement, and Staff Directory.

Optional and Future Scope Features include the AI Assistant (RAG-based), with possible future integration of Learning Management Systems (LMS) and multilingual support.

The exact classification of each feature is defined in Section 8. This Project Scope sub-section describes the boundaries of the system as a whole; Section 8 governs which features are prioritised for initial delivery versus later phases.

1.3.2 Out of Scope

The following are explicitly excluded from the current scope of this SRS:

- Full multilingual support (Sinhala and Tamil), the system is delivered in English only in this phase
- Integration with third party Learning Management Systems (e.g. Google Classroom, Moodle)

1.4 Definitions, Acronyms and Abbreviations

Term	Definition
SRS	Software Requirements Specification — this document
MVP	Minimum Viable Product, the first functional release of the platform
VidyaConnect	The name of the platform. "Vidya" (Sanskrit) means knowledge or education
Super Admin	The highest-privilege system user who manages the entire platform and all schools
School Admin	A school staff member with administrative access to manage one school's data, users, and communications
Teacher	A school staff member who manages class-level communication, assignments, attendance, and events
Parent / Guardian	A parent or legal guardian of a student, with access limited to their own child's information
Student	A pupil enrolled in a school, with read-only access to their own academic data
Class	A group of students assigned to a specific grade or section within a school
Academic Year	The official school year defined by the school admin, within which terms and events are organised
Assignment	A homework task or academic exercise posted by a teacher with a defined deadline
Consent Form	A digital form distributed by a school admin to collect parent approval or information
Announcement	A broadcast message sent school-wide or at class level by an admin or teacher
AI Assistant	The conversational, chat-based feature that allows users to ask natural-language questions and receive answers generated by the system
JWT	JSON Web Token, a standard for secure, stateless authentication

	between client and server
API	Application Programming Interface , the communication layer between the mobile/web client and the backend server
REST	Representational State Transfer — the API architectural style used in this system
GDPR	General Data Protection Regulation — the EU data privacy standard adopted as a best-practice baseline for this project
PaaS	Platform as a Service — the cloud hosting model used for backend deployment
ERD	Entity Relationship Diagram — used to model the database schema

2. Product Overview

2.1 Product Perspective

VidyaConnect is a new, standalone software product. It is not an extension, plugin, or module of any existing school management system, and it does not depend on any pre-existing institutional software. It is being developed independently to directly address the communication and administrative fragmentation currently experienced by Sri Lankan schools.

While VidyaConnect is a self-contained system, it is designed to integrate with a small number of external services to extend its functionality rather than to replace its core logic:

- Firebase Cloud Messaging (FCM) for delivering push notifications to mobile devices.
- Cloud file storage (Supabase Storage or AWS S3) for handling file attachments, consent form uploads, and profile images.

The system consists of two client-facing components and one shared backend:

- A mobile application (React Native), used primarily by parents, students, and teachers

- A web portal (Next.js), used primarily by school administrators and teachers for tasks requiring larger screens, such as bulk data uploads and consent form creation
- A shared backend system (Node.js/Express with PostgreSQL) that serves both clients through a common REST API, ensuring consistent data and business logic across platforms

2.2 Product Vision and Mission

2.2.1 Vision

To become the single trusted communication and administrative platform for every school in Sri Lanka replacing fragmented, unreliable channels with a structured, transparent, and accessible system that strengthens the connection between schools, teachers, parents, and students.

2.2.2 Mission

VidyaConnect's mission is to eliminate the communication gaps that currently cause missed deadlines, lost consent forms, and administrative overload in Sri Lankan schools, by providing:

- A single source of truth for all school related information, replacing scattered WhatsApp groups, printed circulars, and paper forms
- Role aware access, ensuring every user whether a parent, teacher, student, or administrator sees only the information relevant and authorised to them
- A platform built for local realities: low cost Android devices, unstable connectivity, and varying levels of digital literacy
- Transparency and accountability through delivery confirmations, read receipts, and a searchable record of past communications
- Intelligent, time saving tools for administrators and teachers, including bulk data management and an AI powered assistant that can answer questions instantly using the school's own verified records

VidyaConnect is built on the belief that no parent should miss a critical deadline because a message scrolled out of view, and no teacher should spend hours chasing paper forms when a structured digital workflow can do it reliably.

2.3 Product Functions Summary

VidyaConnect provides the following high level functions, grouped by module. Each is described in full functional detail in Section 6.

Module	Function Summary
Authentication and Access Control	Secure, role based login and account management for all user types
Communication Hub	School wide and class level announcements, one-to-one messaging, delivery confirmation, and searchable message archive
Assignment and Task Tracking	Teachers post assignments, parents and students track deadlines and submission status
Smart Notification System	Configurable, intelligent alerts including weekly digests, urgent risk alerts, and event reminders
Digital Consent and Admin Forms	Digital creation, distribution, submission, and tracking of consent and permission forms
Attendance Management	Daily attendance recording by teachers, with visibility for parents and administrators
School Calendar and Event Management	Unified, role aware calendar combining school wide and class specific events
Insights Dashboard	Role specific summary dashboards for parents, teachers, and administrators
Bulk Data Management	Excel based bulk import and export of student, parent, staff, and academic data
School Notice Board	Centralised, browsable archive of official school notices
Student Progress and Academic Records	Structured storage and visibility of student academic performance over time
Parent Teacher Meeting Scheduler	Structured booking and management of parent teacher meeting appointments
School Community and Engagement	Tools supporting broader school community interaction and participation

AI Assistant (RAG-based)	Conversational, natural-language assistant that retrieves and presents accurate, role-authorised data from the platform in response to user questions

2.4 User Classes and Characteristics

VidyaConnect supports five primary user classes, each with distinct access levels, technical characteristics, and usage patterns.

2.4.1 Super Admin

The highest-privilege user, responsible for managing the overall platform and onboarding new schools. Expected to be technically proficient, accessing the system primarily through the web portal.

2.4.2 School Administrator

A school staff member responsible for school-wide configuration, account management, announcements, consent forms, and bulk data operations. Typically moderate-to-high digital literacy; uses the web portal for administrative tasks and may also use the mobile app for monitoring.

2.4.3 Teacher

Responsible for class-level communication, assignment posting, attendance recording, and parent messaging. Digital literacy varies; the system must remain simple and fast to use during and between classes. Primarily uses the mobile app, with web portal access for bulk or detailed tasks.

2.4.4 Parent / Guardian

The largest and most varied user group. Digital literacy ranges from highly proficient to very limited. Many parents use entry-level Android devices with unstable mobile data connections. The interface for this group must be icon-driven, minimal-text, and tolerant of poor connectivity. Parents with multiple children require a unified, switchable view across all their children, who may be enrolled in the same or different classes or schools.

2.4.5 Student

Secondary-level students with read-only access to their own academic data. Assumed to have reasonable familiarity with smartphones but may share devices with parents. Access can be disabled school-wide at the administrator's discretion, and credentials are kept separate from parent accounts for privacy and accountability.

A detailed breakdown of permissions for each role is provided in Section 5 (User Roles and Permissions).

2.5 Operating Environment

VidyaConnect must operate reliably within the following environment:

2.5.1 Mobile Application

- **Operating System:** Android 8.0 and above (primary target); iOS support as a secondary priority
- **Connectivity:** Variable and often unstable mobile data connections (2G/3G in many areas), the application must function with low data usage and support offline access to core features

2.5.2 Web Portal

- **Browsers:** Latest two versions of Chrome, Firefox, Safari, and Edge
- **Devices:** Desktop and laptop computers used primarily by school administrators and teachers

2.5.3 Backend Infrastructure

- **Hosting:** Cloud Platform as a Service (e.g. Railway or Render)
- **Database:** PostgreSQL, hosted in the cloud
- **Notification delivery:** Firebase Cloud Messaging (FCM)
- **File storage:** Supabase Storage or AWS S3 (Section 14, Constraints and Limitations)
- **AI Assistant infrastructure:** A vector database for storing retrievable school data, combined with an external LLM provider API, integrated through the backend's RAG pipeline

2.7 Assumptions and Dependencies

2.7.1 Assumptions

- It is assumed that each school onboarded to the platform will have at least one designated School Administrator with access to a smartphone or computer and basic digital literacy.
- It is assumed that the majority of parent users have access to an Android smartphone (version 8.0 or above) capable of running the mobile application.
- It is assumed that teachers will manually enter assignment, attendance, and academic record data in this phase, as no LMS integration is included in the current scope.
- It is assumed that the free or low cost tiers of third party services (Firebase Cloud Messaging, cloud hosting, file storage) will be sufficient to support a single pilot school deployment of up to 800 students.

- It is assumed that the underlying LLM provider selected for the AI Assistant will support API based integration compatible with a RAG architecture, including the ability to accept retrieved context alongside user queries.

2.7.2 Dependencies

- The system's notification functionality is dependent on the continued availability and free tier sufficiency of Firebase Cloud Messaging.
- The AI Assistant feature is dependent on the platform's core data modules (Communication Hub, Assignment Tracking, Attendance, Consent Forms, Calendar) being live and populated with real data, since the RAG pipeline retrieves directly from this data to ground its responses. The AI Assistant cannot be meaningfully developed or tested in isolation from these modules.
- The AI Assistant is further dependent on the selection of, and successful integration with, a third party LLM provider API, which remains an open item pending mentor feedback (see Section 14).
- Calendar export functionality is dependent on continued compatibility with the standard iCal (.ics) format used by Google Calendar and Apple Calendar.
- The system's hosting and deployment pipeline is dependent on the continued availability of the chosen cloud PaaS provider (Railway or Render) and its free tier terms remaining viable for the pilot deployment phase.

3. Target Users

3.1 Primary Stakeholders

Primary stakeholders are those who directly interact with the VidyaConnect system on a daily basis.

3.1.1.Parents and Guardians

Field	Detail

Who they are	Mothers, fathers, or legal guardians of enrolled students
How they use the system	Receive announcements, track assignments, submit consent forms, communicate with teachers, monitor attendance
Why they matter	They are the main focused user group and the primary audience for school communication
Key concern	Missing important school notices and having no visibility of their child's daily academic life
Access device	Primarily smartphones

3.1.2.Teachers

Field	Detail
Who they are	Class teachers in the school
How they use the system	Post assignments, mark attendance, send announcements, communicate with parents, manage class events
Why they matter	They serve as the **main link** between the school and parents/guardians, facilitating ongoing communication regarding students' academic progress and school activities.
Key concern	Communication overload and the burden of managing multiple informal channels simultaneously
Access device	Mobile app and web portal

3.1.3.School Administrators

Field	Detail
Who they are	Principal, vice principal, and administrative staff responsible for managing the school

How they use the system	Manage accounts, post school-wide notices, create consent forms, configure calendar, generate reports, bulk upload data
Why they matter	They control the platform configuration and are responsible for official school communication
Key concern	No reliable way to track whether official notices were received and acted upon by parents
Access device	Primarily web portal, also mobile app

3.1.4.Students

Field	Detail
Who they are	Enrolled students at the school
How they use the system	View assignments, deadlines, announcements, upload answer sheets, view marks
Why they matter	They are the subject of most communications and benefit from direct visibility of their academic obligations
Key concern	No structured visibility of deadlines and academic responsibilities
Access device	Mobile app (read-only access)

3.1.5 Super Administrator

Field	Detail

Who they are	The platform operator. The VidyaConnect designated technical administrator
How they use the system	Approve and Register schools, manage school admin accounts, configure system-wide settings, monitor platform health
Why they matter	They are responsible for the overall platform operation and onboarding of new schools
Key concern	Ensuring platform stability, data security, and correct school configuration
Access device	Web portal (full system access)

3.2 Secondary Target Users

3.2.1 Industry Mentor — Mr. Ishan Liyanage

Field	Detail
Who he is	The assigned industry mentor guiding the project
Interest in the system	Ensuring the platform is technically sound, realistic, and aligned with real industry standards
How they are affected	Responsible for reviewing scope, providing feedback, and approving major decisions

3.3 Target User Needs Summary

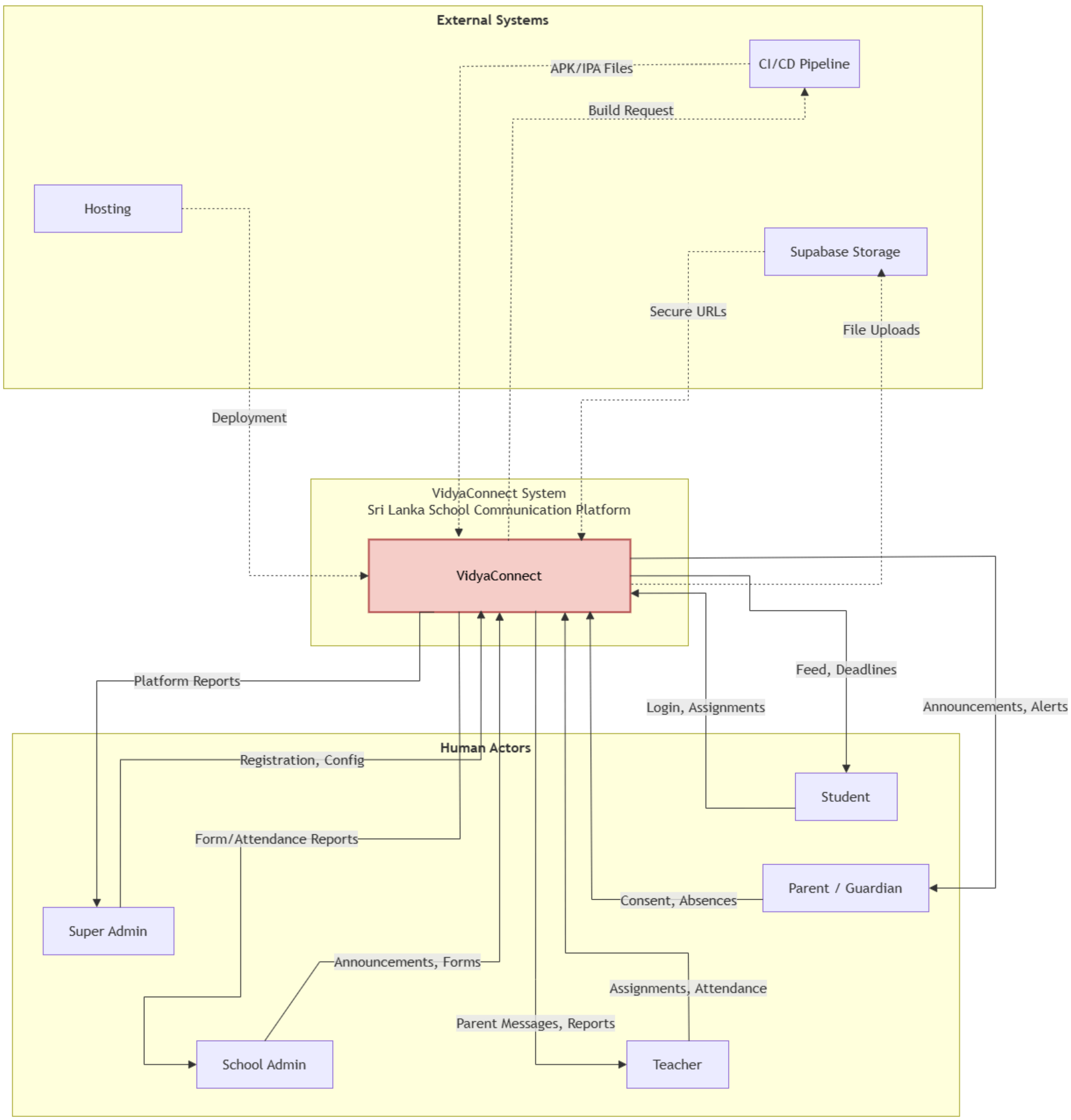
Stakeholder	Primary Need	How VidyaConnect Addresses It

Parent / Guardian	Never miss an important school notice or deadline	Push notifications, weekly digest, announcement feed, consent form alerts, attendance notifications
Teacher	Reduce communication workload without losing reach	Create and manage school communities, post class announcements, automated reminders, message availability hours, delivery confirmation
School Administrator	Manage the entire school communication and set of operations digitally through the platform. Behave like the school.	Broadcast school announcements, create and manage school communities, real-time read receipts, consent form response dashboard, bulk data management, analyze attendance daily
Student	Clear visibility of academic obligations and deadlines	Assignment feed, timetable, deadline countdown, personal calendar view
Super Administrator	Reliable platform operation and easy school onboarding	Broadcast system announcements, schools approve and registration flow, system-wide monitoring
Industry Mentor	Technically sound and realistically scoped project	Clear MVP, extended scope, documented architecture and business rules

4. System Context and Boundaries

4.1 System Context Diagram

https://drive.google.com/file/d/1QrJX6T0Qtn0YtNc3si7cCqB4g6HDw8Zc/view?usp=sharing



4.2 System Boundary Definition

The system boundary defines the functionalities and components that are included within the VidyaConnect system as well as those that are intentionally excluded from the project scope.

4.2.1 Components Included Within the System Boundary

Component	Description
Mobile Application	React Native mobile application used by Parents, Teachers, and Students.
Web Portal	Next.js web portal used by School Administrators and authorized staff.
REST API	Node.js and Express backend responsible for processing system requests and business logic.
Authentication and Authorization	JWT-based authentication and role-based access control mechanisms.
Communication Hub	Management of announcements, class messages, and teacher-parent communication.
Assignment Management	Creation, distribution, tracking, and deadline management of assignments.
Attendance Management	Recording attendance, notifying absences, and managing attendance-related information.
Digital Consent Forms	Creation, distribution, collection, and tracking of parent consent forms.
Event Calendar	Management of school events, holidays, and important academic dates.
Notification System	Delivery of push notifications, reminders, emergency alerts, and weekly summaries.
Insights Dashboard	Role-based dashboards providing summaries, statistics, and reports.
Bulk Data Management	Import and export of school data through spreadsheet files.
Student Progress Tracking	Monitoring assignment completion, attendance trends, and academic performance.

AI Assistant (RAG-based)	Provides intelligent responses to user queries by retrieving relevant information from school documents, announcements, policies, and other authorized knowledge sources before generating answers.
Database	PostgreSQL database used to store structured application data.
File Storage	Supabase Storage used for storing documents, attachments, and uploaded files.
School facility Fee Payment Processing	Requires financial regulations, payment gateway integration, and additional compliance requirements.

4.2.2 Components Excluded from the System Boundary

Component	Reason for Exclusion
Learning Management System (LMS) Integration	Considered a future enhancement and not required for the Minimum Viable Product (MVP).
Multi-language Support	The initial version of the system will be developed in English only. Support for Sinhala, Tamil, and additional languages is planned as a future enhancement.

4.3 External Systems and Integrations

4.3.1 Cloud Hosting

Field	Detail
Purpose	Host the Node.js backend API and PostgreSQL database

Why we chose it	Zero configuration deployment, free tier suitable for MVP, no DevOps overhead for student team
Dependency risk	If the cloud host is unavailable, the entire backend is down. It mitigated by automatic restart and health checks

4.3.2 Supabase Storage

Field	Detail
Purpose	Store and retrieve files such as consent form attachments, absence letters, medical certificates, bulk Excel uploads
What we send to it	Files uploaded by users
What we receive from it	Secure file URLs for retrieval
Why we chose it	Natively integrated with backend, role-based access control, free tier sufficient for pilot
Dependency risk	If Supabase is unavailable, file uploads fail, core text-based features remain functional

5. User Roles and Permissions

5.1 Role Overview

VidyaConnect employs a role-based access control (RBAC) mechanism to ensure that users can only access information and perform actions relevant to their responsibilities. Each user is assigned a predefined role that determines the features and data available to them.

The system supports the following user roles:

- Super Administrator
- School Administrator / Moderator
- Teacher
- Parent / Guardian
- Student

Each role is designed to maintain security, protect sensitive information, and support efficient school communication.

5.2 Super Admin

Description

The Super Administrator is responsible for managing the VidyaConnect platform at the system level. This role is typically assigned to the platform owner or technical operations team.

Responsibilities

- Register and onboard schools.
- Create and manage School Administrator accounts.
- Configure system-wide settings.
- Monitor platform health and performance.
- Manage security policies and access controls.
- View platform-wide analytics and reports.
- Manage AI Assistant (RAG) knowledge sources and system configurations.

Permissions

- Full access to all system modules.
- Create, update, disable, and delete school accounts.
- Manage user roles and permissions.
- Access system logs and audit records.

- Configure notification services.
- Manage storage and database settings.

5.3 School Admin

Description

The School Administrator manages VidyaConnect within a specific school and oversees communication and operational activities.

Responsibilities

- Manage teacher, parent, and student accounts.
- Publish school-wide announcements.
- Manage the school calendar.
- Create and distribute digital consent forms.
- Monitor attendance and assignment statistics.
- Generate school-level reports.
- Moderate inappropriate content and user activities.

Permissions

- Create and manage school users.
- Publish and edit announcements.
- Create school events and calendar entries.
- Generate attendance and engagement reports.
- Manage consent forms.
- Access school-level dashboards.
- Import and export school data.

Restrictions

- Cannot access other schools' data.
- Cannot modify system-wide settings.

5.4 Teacher

Description

Teachers serve as the primary communication link between the school and parents/guardians while managing academic activities for their assigned classes.

Responsibilities

- Post assignments and learning materials.
- Mark and manage attendance.
- Communicate with parents and students.
- Publish class announcements.
- Record student marks and feedback.
- Track student progress.

Permissions

- Create and edit assignments.
- View assigned class lists.
- Mark attendance.
- Send messages to parents and students.
- Upload documents and learning resources.
- View student performance statistics.
- Access AI Assistant (RAG) for retrieving school policies, academic information, and communication support.

Restrictions

- Cannot access administrative settings.
- Cannot modify school-wide configurations.

5.5 Parent / Guardian

Description

Parents and guardians use VidyaConnect to stay informed about their child's academic progress and school activities.

Responsibilities

- Monitor student attendance.
- Track assignments and deadlines.
- Receive school announcements.
- Submit digital consent forms.
- Communicate with teachers.
- Review student performance information.

Permissions

- View child-specific information.
- Receive notifications and reminders.
- Submit consent form responses.
- Communicate with teachers.
- View attendance records.
- View assignment status and academic progress.
- Access AI Assistant (RAG) for answers related to school policies, events, schedules, and frequently asked questions.

Restrictions

- Cannot access information belonging to other students.
- Cannot modify academic records.

5.6 Student

Description

Students use VidyaConnect to stay informed about academic responsibilities, assignments, and school activities.

Responsibilities

- View assignments and deadlines.
- Access announcements.
- Track attendance records.

- Review marks and feedback.
- Submit assignment responses when enabled.

Permissions

- View personal academic information.
- View assignments and deadlines.
- Access school announcements.
- Upload assignment submissions.
- Receive notifications and reminders.
- Access AI Assistant (RAG) for academic guidance and school-related information.

Restrictions

- Cannot modify attendance records.
- Cannot access other students' information.
- Cannot publish announcements.

5.7 Role Permission Matrix

Permission / Feature	Super Admin	School Admin	Teacher	Parent	Student
Manage Schools	✓	✗	✗	✗	✗
Manage School Admin Accounts	✓	✗	✗	✗	✗
Manage Users	✓	✓	✗	✗	✗
View School Dashboard	✓	✓	✓	✗	✗
Publish Announcements	✓	✓	✓	✗	✗
Manage School Calendar	✓	✓	✗	✗	✗
Create Assignments	✗	✗	✓	✗	✗
View Assignments	✓	✓	✓	✓	✓
Submit Assignments	✗	✗	✗	✗	✓
Mark Attendance	✗	✓	✓	✗	✗
View Attendance	✓	✓	✓	✓	✓
Manage Consent Forms	✓	✓	✗	**Submit Only**	**View Only**
Send Messages	✓	✓	✓	✓	**Limited**
View Academic Progress	✓	✓	✓	✓	✓
Generate Reports	✓	✓	✗	✗	✗
Bulk Data Import/Export	✓	✓	✗	✗	✗
Access AI Assistant (RAG)	✓	✓	✓	✓	✓
Configure System Settings	✓	✗	✗	✗	✗
Access Audit Logs	✓	✗	✗	✗	✗

6. Functional Requirements

6.1 Authentication and Access Control

6.1.1 User Registration and Login

- 5 user roles; no self-registration; Super Admin creates School Admin accounts
- Login via username/email + password; JWT token issued
- First-time users must change temporary password

6.1.2 Role-Based Access Control

- Role permissions enforced at API and UI layers; each role has strict boundaries
- Super Admin: full platform; Teachers: assigned classes only; Parents: linked child only
- 403 Forbidden on any cross-role/cross-school access
- Database queries include mandatory school-scope filtering

6.1.3 Password Management

- Passwords hashed with bcrypt (cost 12); 8 chars min with complexity requirements
- Forgot Password: 15-min expiry email link (single-use)
- Cannot reuse last 5 passwords
- Admins can force password reset for any user
- Access tokens: 24-hour expiry; refresh tokens: 7-day expiry
- Auto-refresh while active; invalidate on logout or role change
- Inactive sessions (30+ min): require re-authentication
- Concurrent sessions supported on multiple devices

6.1.4 Multi-Child Parent Account

- Single parent account links to multiple children (max 2 parents per child)
- Parent switches child profiles without logging out
- Notifications/assignments/forms filtered by selected child
- Linking requires School Admin approval

6.2 Communication Hub

6.2.1 School-Wide Announcements

- Admins create announcements visible to all school users
- Support rich text, hyperlinks, file attachments (≤10 MB); push notification on publish
- Can schedule, edit, or retract; retracted shows placeholder
- Admins view read receipt counts and percentages

6.2.2 Class-Level Messages

- Teachers message assigned classes; visible to parents of that class
- Support text, files (≤10 MB), images; can send to multiple classes simultaneously
- Teachers can pin messages for defined duration
- Appears in Communication Hub feed with class name and teacher name

6.2.3 Teacher-Parent Private Messaging

- One-to-one between teacher and parents of their students only
- Conversational interface with timestamps; all messages logged (admin accessible for safeguarding)
- Teachers cannot message parents of other classes
- Students cannot initiate direct messages

6.2.4 Delivery Confirmation and Read Receipts

- System tracks: Sent → Delivered → Read status
- Read receipts show count/percentage; private messages show individual status
- Admin views detailed read report per announcement
- Parents cannot disable read receipts from admin/teacher

6.2.5 School Community Groups

- Admins create community groups with role-based or open membership
- Group messages: rich text, files, attachments, read receipts
- Support pinned messages and group descriptions
- Moderated by admin

6.3 Assignment and Task Tracking

6.3.1 Assignment Creation by Teacher

- Teachers create with: title, subject, description, class(es), due date/time, attachments
- Can save as draft, duplicate as template, edit after publish
- Publishing notifies students/parents; cancellations trigger notifications

6.3.2 Assignment Feed for Parents

- Consolidated feed shows all active assignments for linked child
- Display: subject, title, teacher, due date, time remaining, submission status
- Filterable by subject, status, due date; overdue highlighted
- 24-hour pre-deadline notification

6.3.3 Assignment Feed for Students

- Personalised feed of all assignments sorted by nearest deadline
- View details, download files, mark "In Progress", see submission status
- Completed/past assignments in separate section
- Notifications for new assignments, deadlines, feedback

6.3.4 Submission Status Tracking

- Four states: Not Submitted, Submitted (on-time), Late, Missing
- Real-time updates on teacher dashboard and parent/student feeds
- Teacher can manually override status
- Submission summary shows count/percentage per status

6.3.5 Teacher Marking and Feedback

- Upload confirmation + status updates to "Submitted"
- Cannot resubmit after deadline unless teacher enables

6.3.7 Automated Deadline Reminders

- Notification 24 hours and 1 hour before deadline (if not submitted)
- Teachers can disable per assignment
- All reminders logged in notification history

6.3.8 Digital Calendar View

- Calendar-style layout showing assignments per day (colour-coded by subject)
- Parent views current week + navigate forward/backward
- Offline accessible if loaded in past 24 hours
- Available to students and parents

6.4 Smart Notification System

6.4.1 Push Notification Delivery

- Delivered via FCM within 60 seconds of event
- Each notification includes title, body, timestamp, deep link
- Fallback to in-app notification log if FCM unavailable

6.4.2 Risk Alerts

- Triggered: 2+ overdue assignments, unread announcement >48 hrs
- Sent to parent + flagged on teacher dashboard
- Distinct warning appearance
- Requires teacher/admin acknowledgement

6.4.4 Event Reminders

- Auto-sent 3 days and 1 day before events
- Include title, date, time, location, event details link

6.4.5 Emergency Alerts

- Bypasses all notification preferences; delivered immediately
- Can broadcast to entire school or selected classes
- Distinct visual/audible treatment; parents cannot disable
- Logged permanently in audit

6.4.6 Notification Preferences

- Users configure per notification type (announcements, assignments, reminders, etc.)

- Emergency alerts always enabled (non-configurable)
- Admin can define mandatory notification types school-wide
- Stored per-user per-device

6.4.7 Notification Log and History

- In-app log shows all notifications in reverse chronological order
- Each entry: title, body, timestamp, read/unread status, type
- Retain 90 days per user
- Mark all as read; tap to navigate to relevant screen

6.4.8 Parent Reading Confirmation

- Announcements require explicit "Confirm Read" action
- Announcement flagged "Action Required" until confirmed
- Admin views confirmed vs. non-confirmed list in real-time
- Auto-reminders every 24 hours until confirmed

6.5 Digital Consent and Admin Forms

6.5.1 Form Creation by Admin

- Support field types: text, long text, single/multi-choice, yes/no, date, number, file upload
- Mark fields required/optional; set submission deadline
- Add title, description, cover image; save as draft or duplicate
- Export submissions as Excel with summary sheet

6.5.2 Form Distribution

- Distribute to: all parents, specific class/grade, individual accounts
- Targeted parents get push notification
- Admin views distribution list anytime
- Cannot distribute to inactive accounts

6.5.3 Parent Form Submission

- Parents view forms in dedicated Forms section with deadline and status

- Complete and submit in-app; file uploads (PDF, JPEG, PNG; ≤10 MB)
- Confirmation notification on submit; cannot edit unless admin re-opens
- Admin receives notification on submission

6.5.4 Digital Acknowledgement

- Formal acknowledgements (policy, fees): explicit statement + checkbox required
- Timestamp and parent identity recorded permanently
- Exportable for legal/audit record-keeping

6.5.5 Real-Time Response Dashboard

- Shows: total recipients, submitted %, non-submitted %, response breakdown per field
- Updates in real-time; drill-down to individual parent responses
- Export data as Excel with summary statistics

6.5.6 Automated Reminders

- Auto-remind at: 3 days, 1 day, deadline day before due
- Admin can configure/disable per form
- Admin can manually trigger extra reminder blast

6.5.7 Form Archive and Export

- All responses archived minimum 3 years
- Export as Excel with timestamps per parent
- Searchable by date, form type, parent name

6.5.8 Fee Payment Acknowledgement

- Admin attaches fee notice, parent confirms receipt
- Outstanding acknowledgements flagged on admin dashboard

6.6 Attendance Management

6.6.1 Daily Attendance Marking by Teacher

- Teachers mark attendance from mobile/web with alphabetical student list
- Mark per-period (morning/afternoon) or single daily (configurable)
- Once submitted: locked unless teacher requests admin amendment
- Reminder sent if not marked by deadline (e.g., 9:30 AM)

6.6.2 Attendance Status Types

- Status values: Present, Absent, Absent–Excused, Late.
- Status displayed in dashboard and parent notification

6.6.3 Attendance Notification to Parent

- Auto-push sent within 5 min of "Absent" recorded
- Includes student name, date, prompt to submit justification
- Direct link to justification submission

6.6.4 Parent Absence Justification

- form: reason type (Illness/Emergency/Pre-Arranged/Other), text, optional document
- Submittable up to 3 days after absence
- Status updates to "Absent–Justified (Pending Review)"
- Teacher accepts (→ "Absent–Excused") or rejects with note; parent notified

6.6.5 Medical Certificate Upload

- For 3+ consecutive illness absences: parent prompted to upload certificate
- Accept PDF or image (JPEG, PNG; ≤10 MB)
- Secure storage; accessible only to teacher + admin
- Dashboard flags multi-day absences requiring certificate

6.6.6 Attendance History View

- Teacher: full history per student (filterable by month/status)
- Parent: child's history with statuses, justifications, outcomes
- Student: own attendance summary
- Visual summary: calendar heatmap or monthly bar chart

6.6.7 Monthly Attendance Report

- Auto-generated first school day each month (for previous month)
- Includes: total school days, % per student, absence/late counts, class average
- Exportable as PDF or Excel
- Accessible to teacher and admin

6.6.8 Low Attendance Alert

- Triggered <80% attendance in any month
- Sent to parent
- Includes current % and absence count

6.7 School Calendar and Event Management

6.7.1 School-Wide Event Creation

- Admin creates events: title, date, time (or all-day), description, category, attachments
- Categories: Academic, Sports, Cultural, Administrative, Holiday
- All users notified on creation; admin can edit/cancel

6.7.2 Class-Level Event Creation

- Teachers create events for assigned classes only
- Same fields as school-wide; scope restricted to selected classes

6.7.3 Role-Based Calendar View

- Admin: all events; Teacher: school + own class events; Parent: school + child's class; Student: school + own class

6.7.4 Conflict Detection

- System checks for conflicts with existing events on same date/time
- Warning displayed; user can proceed if desired

6.7.5 Holiday Blocking

- Admin defines public holidays and school closures
- Visually marked in all user calendars
- Warning if creating event/assignment on holiday (does not prevent)

6.7.6 Event RSVP

- Admin/teacher enables RSVP (Yes/No/Maybe)
- Dashboard shows count per response + list of responders
- Auto-reminders 5 and 2 days before for non-responders
- Parents change RSVP up to 24 hours before event

6.7.7 Calendar Export

- Users export events to personal calendar
- One-tap export per event
- Bulk export all upcoming events option

6.7.8 Exam Timetable and Countdown

- Admin publishes exam timetable as structured events (subject, date, time, class)
- Prominently displayed with countdown in student/parent calendars
- Not editable by teachers; only admin can modify
- Notifications on publish and 3 days before each exam

6.8 Insights Dashboard

6.8.1 Parent Dashboard

- Personalised summary: pending assignments + deadline, attendance %, unread announcements, outstanding forms, upcoming events (7 days)
- Auto-refreshes on app open
- Each card tappable → navigates to module

6.8.2 Teacher Dashboard

- Class-level summary: today's attendance %, pending submissions/marking, unread parent messages, form response %, upcoming events
- Multi-class aggregated view with filter-by-class

6.8.3 Admin Dashboard

- School-wide overview: today's attendance %, active announcements + avg read %, outstanding forms, low-attendance students, upcoming events (14 days), pending account setups
- Visual cards with trend indicators (up/down vs. last week)

6.8.4 Student Progress Tracker

- Shows: avg mark, subject-wise avg, trend direction (improving/declining/stable), recent marks list
- Only displays marked/returned assignments
- Visible to teacher, parent, student

6.8.5 Assignment Completion Rate

- % of students who submitted on-time per assignment per class
- Historical trends (weekly/monthly)
- Drill-down to per-student completion rates

6.8.6 Attendance Trend View

- Line/bar chart of daily/weekly attendance rates for current term
- Significant dips highlighted automatically
- Admin can drill-down to class-level data

6.9 Bulk Data Management

6.9.1 Student Bulk Upload

- Admin uploads Excel template: Full Name, DOB, Class, Gender, Student ID (optional)
- System validates all rows; rejects if errors; provides error report
- Auto-creates accounts with system-generated temp password
- Successful imports immediately usable

6.9.2 Parent and Staff Bulk Upload

- Separate templates for parents (Name, Email, Phone, Student IDs) and staff (Name, Email, Role, Assigned Classes)
- System auto-links parents to students during upload
- Duplicate detection flags existing emails

6.9.3 Login Bulk Creation

- Bulk upload auto-generates credentials (username + temp password)
- Admin downloads credential sheet (Excel) for distribution
- Generated once only; password reset if credentials lost

6.9.4 Exam Marks Bulk Import

- Teacher/admin uploads Excel: Student Name, ID, Subject, Exam, Mark
- System validates marks against max mark per exam
- Imported marks immediately visible in Student Progress Tracker

6.9.5 Upload Validation and Error Reporting

- Pre-commit validation: missing fields, invalid formats, duplicates, non-existent references
- Detailed error report (Excel) with row #, column, reason per error
- Entire batch must pass; no partial commits

6.9.6 Bulk Data Export

- Export student list, parent list, staff, attendance, forms, marks as Excel
- Cover sheet: school name, export date, date range, record count
- Large datasets processed asynchronously; admin notified when ready

6.10 School Notice Board

6.10.1 Digital Notice Posting

- Admin posts notices: rich text, images, attachments (≤10 MB)
- Display: title, category, posting date, admin name
- All users notified on posting

6.10.2 Notice Categories

- Pre-defined: Academic, Administrative, Event, Health & Safety, Financial, Sports, General
- Users filter by category
- Admin can add/rename from settings

6.10.3 Notice Expiry

- Admin sets expiry date; auto-removed after (default 30 days)
- Admin can extend/remove expiry
- Expired notices move to archive

6.10.4 School Circular Archive

- All past notices in searchable archive (≥3 years)
- Search by keyword, category, date range, author
- All users can access; admin can reinstate to active board

6.10.5 Morning Assembly Notices

- Teachers/admins post short, time-sensitive morning announcements
- Highlighted at top of notice board each morning
- Auto-expire end of school day (5:00 PM) → archive

6.11 Student Progress and Academic Records

6.11.1 Progress Report Distribution

- Admin/teacher upload term or mid-term reports as PDF per student
- Upon upload: parent + student notified + can view/download from app
- Reports immutable once published; corrected version → new upload

6.11.2 Subject-Wise Performance View

- Display: subject name, teacher, # assessments, avg mark, highest mark, trend
- Visible to teacher, parent, student
- Only shows marked/returned assignments

6.11.3 Term Report Upload and Access

- Bulk upload: zip of PDFs named with pattern (StudentID_TermReport_Term1.pdf)
- System auto-matches PDF to student via filename
- Reports accessible to student + parents for account lifetime

6.11.4 Student Achievement Recognition

- Teacher/admin post achievement badges or messages (e.g., Best in Science, 100% Attendance)
- Student + parent notified
- Visible on student profile
- Celebratory/positive reinforcement tone

6.11.5 Class Timetable Viewer

- Admin publishes weekly timetable per class
- Student + parent view current week; teacher views own timetable across classes
- Editable by admin; shows effective date of current version

6.12 Parent-Teacher Meeting Scheduler

6.12.1 Teacher Availability Slot Management

- Teachers define: date, start/end time, duration (e.g., 15 min), max slots

- System auto-divides into bookable slots
- Teachers open/close booking for specific slots or entire window
- Booked slots auto-locked

6.12.2 Parent Meeting Booking

- Parent views available slots per teacher + books one per cycle
- First-come, first-served; once taken, unavailable
- Confirmation to parent + teacher
- Parent can cancel up to 2 hours before meeting

6.12.3 Meeting Confirmation and Reminders

- Confirmation notification on booking to parent + teacher
- Auto-reminder 1 day and 1 hour before
- Cancellation notification sent; slot released for re-booking

6.12.4 Meeting Notes by Teacher

- Post-meeting: teacher records brief notes (plain text, ≤500 words)
- Private to teacher + admin; not visible to parent/student
- Archived; accessible to admin for safeguarding

6.13 School Community and Engagement

6.13.1 School News Feed

- Admin posts news (rich text, images, attachments) to community feed
- All school users see; can react with emojis
- Reverse chronological; no auto-expiry unless manually removed

6.13.2 School Poll and Survey

- Admin/teacher create with: single-choice, multi-choice, text response
- Defined open/close date
- Admin views real-time results
- Parents/students see own response; anonymous polls show aggregate only

6.13.3 Anonymous Suggestion Box

- Anonymous submission (no sender identification)
- Submissions to admin dashboard
- Admin marks reviewed/responded/dismissed + can post public response

6.13.4 Parent Volunteer Coordination

- Admin posts opportunities (date, time, description, needed)
- Parents sign up from app
- Admin confirms/declines; volunteer roster viewable

6.13.5 After School Activity Registration

- Admin posts activities (description, schedule, eligibility, deadline)
- Parent registers child in-app
- Max participant limit enforced; admin downloads roster

7. Non-Functional Requirements

7.1 Performance Requirements

7.1.1 Response Time

- 95% of API requests respond ≤2 seconds (500 concurrent users)
- Dashboard load ≤3 seconds on 4G mobile
- Search results ≤2 seconds for 10k records
- Report generation ≤30 seconds for 5k records (async for larger)

7.1.2 Load Capacity

- Support ≥500 concurrent users per school
- Support ≥10 schools simultaneously
- Peak load (morning attendance): <1% error rate
- Load testing validated before MVP

7.1.3 Data Usage per Page Load

- Mobile screen: ≤500 KB per load (supports limited data plans)
- Images/thumbnails lazy-loaded; full-res on-demand
- Local cache: dashboard, announcements, class lists
- Low-data mode: suppress inline images, reduce notification payload

7.1.4 Notification Delivery Time

- Push notifications: within 60 seconds of event
- Emergency alerts: within 30 seconds
- Bulk notifications (>500): all within 5 minutes
- Monitor delivery rate; alert if <95% success

7.2 Security Requirements

7.2.1 Data Encryption at Rest and in Transit

- Client-server: TLS 1.2+ (HTTP → HTTPS redirect)
- Database PII: AES-256 encryption at rest
- Uploaded files: encrypted in Supabase Storage with signed time-limited URLs
- Backups: encrypted with same standard

7.2.2 Authentication and Authorisation

- All API endpoints require valid JWT in Authorization header
- JWT signed with RS256 (asymmetric)
- Validate: token expiry, signature, issuer on every request
- Authorisation: API layer enforces role + school affiliation per resource

7.2.3 Role Isolation

- One school's data never accessible to other schools
- Cross-tenant/cross-role access → 403 Forbidden + logged as security event
- Database queries include mandatory school-scope filtering

7.2.4 Audit Logging

- Events logged: login/logout, failed login, password change, account creation/deactivation, role change, exports, bulk uploads, form submission, emergency alerts
- Each entry: event type, user ID/role, timestamp (UTC), source IP, affected resource
- Immutable; no deletion/editing allowed
- Retain ≥5 years; admin views school logs, Super Admin views system logs

7.3 Reliability and Availability

7.3.1 System Uptime

- Maintenance: weekday evenings or weekends; 48-hour notice to admins
- Health monitoring: auto-alert if uptime drops below target
- Critical services prioritised in partial degradation

7.3.2 Error Handling and Recovery

- Server errors: user-friendly messages (no stack traces)
- Network failures on form/file upload: auto-retry up to 3 times
- Failed submissions: preserved in local storage
- Third-party integrations: circuit breaker pattern
- Critical errors: auto-logged + alert dev team

7.3.3 Data Backup and Retention

- PostgreSQL backed up every 24 hours (30-day retention)
- Point-in-time recovery (PITR) enabled (last 7 days)
- Backups stored geographically separate
- Restoration tested ≥once per academic term
- Supabase Storage files included in backups

7.4 Usability Requirements

7.4.1 Device Compatibility

- Mobile: Android , iOS
- Web: latest 2 major versions Chrome, Firefox, Safari, Edge

7.4.2 Icon-Driven Navigation

- Bottom nav bar: icon + label pairs for primary modules
- All icons have visible text or tooltip
- Consistent icon library throughout
- Navigation reviewed with representative school users

7.5 Scalability Requirements

7.5.1 User Capacity per School

- MVP: ≤800 students, ≤60 teachers, ≤1,600 parents
- Architecture scalable to 3,000 students without re-architecture
- Performance benchmarks validated before pilot

7.5.2 Multi-School Support

- Multi-tenant by design; each school isolated with strict data segregation
- Onboarding new schools: no code changes or downtime
- Super Admin onboards via platform interface

7.5.3 Storage Scalability

- Cloud object storage (Supabase): auto-scaling
- Storage quota: configurable per school by Super Admin

7.6 Maintainability Requirements

7.6.1 Code Documentation Standards

- README per repo: setup, env config, test/deploy instructions

- Documentation reviewed in PR process

7.6.2 Version Control Requirements

- Git Flow: main (prod), develop (integration), feature /branches
- PRs require ≥1 peer review before merge to develop
- Commits: Conventional Commits format
- No direct commits to main/develop
- .gitignore: exclude secrets, node_modules, artifacts, local env

7.6.3 Deployment and Rollback

- CI/CD: GitHub Actions (tests + linting on PR to develop)
- Merges to main: auto-deploy to Render
- Deployments tagged with semantic version
- Failed deployment: auto-rollback to last stable
- Deployment log recorded

8. Feature Scope Classification

8.1 Core MVP Features

Features delivered by the end of Semester 1 and validated with pilot school.

	Feature	**Justification**
1	Communication Hub	Replaces WhatsApp
2	Assignment Tracking	Daily parent need
3	Smart Notifications	Eliminates spam problem
4	Digital Consent Forms	Replaces paper forms
5	School Calendar	Replaces scattered date announcements
6	Attendance Management	Daily safety and parent communication need

7	Insights Dashboard	Unified view per role
8	Bulk Data Management	Essential for school onboarding
9	Authentication and RBAC	Foundation of the entire system

8.2 Extended Scope Features

	Feature	**Justification**
1	Student Submission and Marking	Closes the assignment loop
2	Student Progress Tracker	High parent value, connects existing data
3	Parent-Teacher Meeting Scheduler	Reduces phone call dependency
4	Progress Report Distribution	Replaces printed report cards
5	Exam Timetable and Countdown	High parent demand
6	School Notice Board and Circular Archive	Replaces physical notice board
7	Emergency Alert System	Safety critical
8	School Community Groups	Extends communication hub
9	Student Achievement Recognition	Positive engagement
10	Class Timetable Viewer	Parent visibility
11	Health and Safety Alert	Public health communication
12	School News Feed	Community building

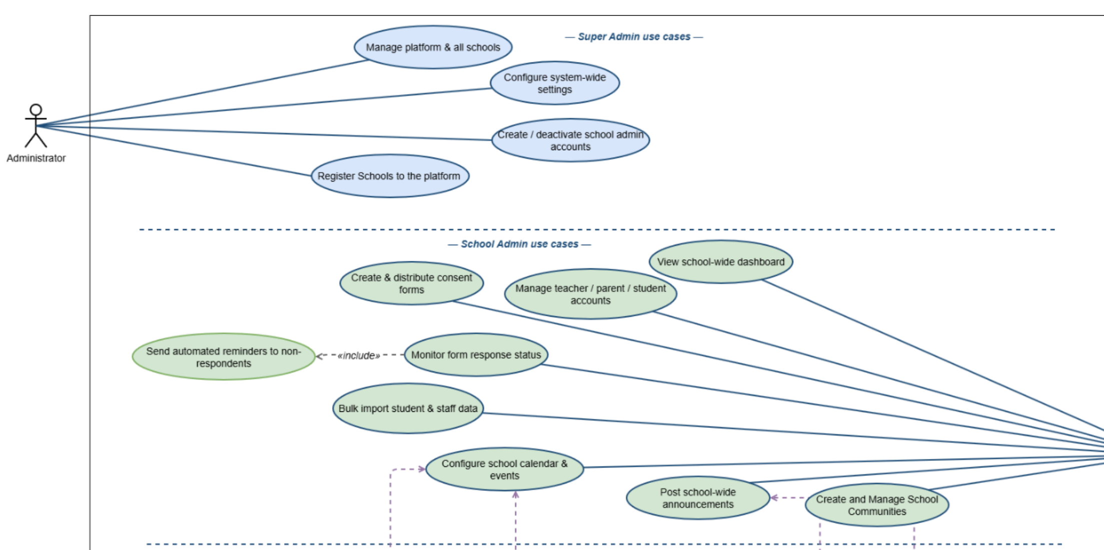
8.3 Optional and Future Scope

	Feature	Reason Deferred
1	AI Assistant	Requires stable data and API cost management
2	Library Management Module	Separate use case
3	Alumni Module	Needs matured user base
4	LMS Integration	Manual entry sufficient for MVP
5	Multilingual Support	English sufficient for pilot
6	Canteen Menu and Pre-Order	Low priority for communication gap
7	School Bus Tracking	Requires coordination infrastructure
8	Anonymous Suggestion Box	Post-MVP engagement feature
9	Parent Volunteer Coordination	Low priority
10	Lost and Found Board	Low priority

9. Use Case Specifications

9.1 Use Case Diagram — Full System

https://drive.google.com/file/d/1PQLSaA-2pgKYf0P0f6xIAEATjoyjj0yj/view?usp=sharing



9.2 Detailed Use Case Descriptions per Module

9.2.1 Module 1 — Authentication and Access Control

UC-01: User Login

Field	Detail
Use Case ID	UC-01
Use Case Name	User Login
Actor	All roles
Description	Any registered user logs into the system using their credentials and gains access based on their assigned role
Precondition	User must have a registered account created by Super Admin or School Admin
Postcondition	User is authenticated and redirected to their role-specific dashboard

UC-02: Password Reset

Field	Detail
Use Case ID	UC-02

Use Case Name	Password Reset
Actor	All roles
Description	User resets their forgotten password through a secure email link
Precondition	User must have a registered email address in the system
Postcondition	User successfully sets a new password and can log in

9.2.2 Module 2 — Communication Hub

UC-03: Post School-Wide Announcement

Field	Detail
Use Case ID	UC-03
Use Case Name	Post School-Wide Announcement
Actor	School Admin
Description	School Admin posts an official announcement visible to all parents, teachers, and students in the school
Precondition	Actor must be logged in as School Admin or Moderator
Postcondition	Announcement is published and all relevant users receive a push notification

UC-04: Send Class-Level Message

Field	**Detail**
Use Case ID	UC-04
Use Case Name	Send Class-Level Message

Actor	Teacher
Description	Teacher sends a message to all parents of students in their assigned class
Precondition	Teacher must be assigned to the class they are messaging
Postcondition	All parents of that class receive the message and push notification

UC-05: Send Private Message to Individual Parent

Field	Detail
Use Case ID	UC-05
Use Case Name	Send Private Message to Individual Parent
Actor	Teacher
Description	Teacher opens a private one-to-one message thread with a specific parent to discuss their child's individual matter
Precondition	Teacher must be assigned to a class containing the parent's child
Postcondition	A private message thread is created between the teacher and parent

UC-06: Create and Manage School Communities

Field	Detail
Use Case ID	UC-06
Use Case Name	Create and Manage School Communities
Actor	School Admin, School Moderator, Teacher

Description	Admin or teacher creates a community group for a specific audience — for example PTA Members, Sports Team Parents, Grade 8 Parents — and posts targeted messages to that group
Precondition	Actor must be School Admin, Moderator, or assigned Teacher
Postcondition	Community is created and relevant members can view community messages

9.2.3 Module 3 — Assignment and Task Tracking

UC-07: Post Assignment to Class

Field	Detail
Use Case ID	UC-07
Use Case Name	Post Assignment to Class
Actor	Teacher
Description	Teacher creates and posts a new assignment for their class with all relevant details
Precondition	Teacher must be logged in and assigned to the class
Postcondition	Assignment is visible to all parents and students of that class and a notification is sent

UC-08: Add Assignment Marks

Field	Detail
Use Case ID	UC-08

Use Case Name	Add Assignment Marks
Actor	Teacher
Description	Teacher adds marks and optional feedback to a submitted assignment
Precondition	Student must have uploaded an answer sheet for the assignment
Postcondition	Mark is saved and parent and student are notified that marks are available

UC-09: View Child's Assignment Feed

Field	Detail
Use Case ID	UC-09
Use Case Name	View Child's Assignment Feed
Actor	Parent / Guardian
Description	Parent views all assignments for their child across all subjects in one organised feed
Precondition	Parent must be logged in and linked to the student
Postcondition	Parent sees full assignment list with status for their child

9.2.4 Module 4 — Smart Notification System

UC-10: Configure Notification Preferences

Field	Detail
Use Case ID	UC-10

Use Case Name	Configure Notification Preferences
Actor	Parent / Guardian
Description	Parent customises how and when they receive notifications from the platform
Precondition	Parent must be logged in
Postcondition	Notification preferences are saved and applied immediately

UC-11: Receive and View Notification Log

Field	Detail
Use Case ID	UC-11
Use Case Name	Receive and View Notification Log
Actor	Parent / Guardian, Teacher
Description	User views a complete history of all notifications they have received
Precondition	User must be logged in
Postcondition	User sees full notification history with timestamps

9.2.5 Module 5 — Digital Consent and Admin Forms

UC-12: Create and Distribute Consent Form

Field	Detail

Use Case ID	UC-12
Use Case Name	Create and Distribute Consent Form
Actor	School Admin
Description	School Admin creates a digital consent or permission form and sends it to selected parents with a submission deadline
Precondition	Admin must be logged in as School Admin
Postcondition	Form is distributed to relevant parents and a notification is sent

UC-13: Monitor Form Response Status

Field	Detail
Use Case ID	UC-13
Use Case Name	Monitor Form Response Status
Actor	School Admin
Description	Admin views real-time response status for a distributed consent form
Precondition	Form must have been distributed to parents
Postcondition	Admin sees live count of submitted, pending, and overdue responses

UC-14: Submit Consent Form Digitally

Field	Detail
Use Case ID	UC-14
Use Case Name	Submit Consent Form Digitally
Actor	Parent / Guardian
Description	Parent receives a consent form, fills it in, and submits it digitally through the app

Precondition	Parent must have received the form and deadline must not have passed
Postcondition	Form response is saved and admin is notified of the submission

9.2.6 Module 6 — Attendance Management

UC-15: Mark Daily Attendance

Field	Detail
Use Case ID	UC-15
Use Case Name	Mark Daily Attendance
Actor	Teacher
Description	Teacher marks attendance for every student in their class at the start of each school day
Precondition	Teacher must be assigned to the class and attendance must not have been marked yet today
Postcondition	Attendance is saved for all students and absent students' parents are notified

UC-16: Submit Absence Justification

Field	Detail
Use Case ID	UC-16
Use Case Name	Submit Absence Justification
Actor	Parent / Guardian
Description	Parent receives an absence notification and submits a reason with optional supporting document
Precondition	Student must have been marked absent and parent must have received the notification

Postcondition	Justification is saved and teacher can view the reason

9.2.7 Module 7 — Calendar and Event Management

UC-17: Configure School Calendar and Events

Field	Detail
Use Case ID	UC-17
Use Case Name	Configure School Calendar and Events
Actor	School Admin
Description	Admin sets up the school calendar by adding term dates, school holidays, and school-wide events
Precondition	Admin must be logged in
Postcondition	Calendar is configured and all users can view school events

UC-18: Add Class Events to Calendar

Field	Detail
Use Case ID	UC-18
Use Case Name	Add Class Events to Calendar
Actor	Teacher
Description	Teacher adds a class-specific event — test, project deadline, field trip — to the calendar for their class
Precondition	Teacher must be assigned to the class and the date must not be a school holiday
Postcondition	Event appears on the calendar for all parents and students of that class

9.2.8 Module 8 — Insights Dashboard

UC-19: View School-Wide Dashboard

Field	Detail
Use Case ID	UC-19
Use Case Name	View School-Wide Dashboard
Actor	School Admin
Description	Admin views a summary of all school activity — announcements, forms, attendance, and assignment completion — in one dashboard
Precondition	Admin must be logged in
Postcondition	Admin sees live school-wide summary

UC-20: View Class Dashboard and Reports

Field	Detail
Use Case ID	UC-20
Use Case Name	View Class Dashboard and Reports
Actor	Teacher
Description	Teacher views a summary of their class performance — assignment completion, attendance, and parent engagement
Precondition	Teacher must be logged in and assigned to a class
Postcondition	Teacher sees class-level summary and can identify students at risk

9.2.9 Module 9 — Bulk Data Management

UC-21: Bulk Import Student and Staff Data

Field	Detail
Use Case ID	UC-21
Use Case Name	Bulk Import Student and Staff Data
Actor	School Admin
Description	Admin uploads a pre-formatted Excel file to create multiple student, parent, or staff records at once — used primarily during school onboarding
Precondition	Admin must have a correctly formatted Excel file using the system template
Postcondition	All valid records are imported and corresponding accounts are created

UC-22: Bulk Export School Data

Field	Detail
Use Case ID	UC-22
Use Case Name	Bulk Export School Data
Actor	School Admin, Super Admin
Description	Admin exports any module's data as a formatted Excel file for school records or reporting
Precondition	Admin must be logged in with export permissions
Postcondition	Excel file is downloaded containing the requested data

9.2.10 Module 10 — Student Progress and Academic Records

UC-23: View Assignment Submission Status

Field	Detail
Use Case ID	UC-23
Use Case Name	View Assignment Submission Status
Actor	Teacher
Description	Teacher views which students have submitted, which are late, and which are missing for a specific assignment
Precondition	Assignment must have been posted
Postcondition	Teacher sees full submission breakdown for the assignment

UC-24: View Assignment Marks

Field	Detail
Use Case ID	UC-24
Use Case Name	View Assignment Marks
Actor	Student, Parent / Guardian
Description	Student or parent views marks and teacher feedback for a completed and marked assignment
Precondition	Teacher must have released marks for the assignment
Postcondition	Student and parent see mark and feedback

10. System Architecture

10.1 Architecture Overview

VidyaConnect follows a **Layered Architecture Pattern**, which separates the system into multiple layers, each with a specific responsibility. This architecture improves maintainability, scalability, security, and ease of future enhancements.

The architecture consists of four primary layers:

1. Client Layer
2. API and Controller Layer
3. Service and Business Logic Layer
4. Data and External Services Layer

User requests originate from the client applications and pass through the API layer, where they are validated and routed to the appropriate business services. The service layer executes business rules and interacts with the database and external services before returning a response to the client.

Architectural Benefits

- Clear separation of responsibilities
- Improved maintainability
- Enhanced security through centralized authentication
- Scalability for future growth
- Easier testing and deployment

10.2 Client Layer

	Mobile Application	Web Administration Portal
Technology	React Native	Next.js
Users	● Parents/Guardians ● Teachers ● Students	● School Administrators ● Super Administrators ● Teachers ●
Key Functions	● View announcements ● Manage assignments ● View attendance ● Submit consent forms ● Receive notifications ● Access AI Assistant	● User management ● School administration ● Reporting and analytics ● Calendar management ● Bulk data operations

Both applications communicate with the **same backend API** through secure **HTTPS** requests.

10.3 API and Controller Layer

The API and Controller Layer serves as the entry point for all system requests.

Technology: Node.js and Express.js

Responsibilities

- Receive client requests
- Validate input data
- Authenticate users

- Authorize user actions using role-based access control (RBAC)
- Route requests to the appropriate service
- Return responses in JSON format

Security Mechanisms

- JWT-based authentication
- Role-based access control
- HTTPS communication
- Input validation and sanitization

This layer ensures that only authorized users can access system resources.

10.4 Service and Business Logic Layer

The Service Layer contains the core business functionality of VidyaConnect.

Service	Responsibility
User Management Service	Manage users, schools, classes, and roles
Announcement Service	Handle school and class communications
Assignment Service	Manage assignments, submissions, and feedback
Attendance Service	Record attendance and process absence notifications
Consent Form Service	Create, distribute, and track digital consent forms
Calendar Service	Manage school events and academic schedules
Notification Service	Deliver push notifications and reminders
Report Service	Generate dashboards and analytics
AI Assistant Service (RAG)	Retrieve school information and generate intelligent responses

10.5 Data and External Services Layer

This layer manages persistent storage and integrations with external platforms.

10.5.1 PostgreSQL Database

Stores:

- User accounts
- School information
- Assignments
- Attendance records
- Announcements
- Consent forms
- Calendar events
- Reports and logs

10.5.2 Supabase Storage

Stores:

- Assignment files
- Supporting documents
- Consent form attachments
- Images and uploaded resources

RAG Knowledge Base

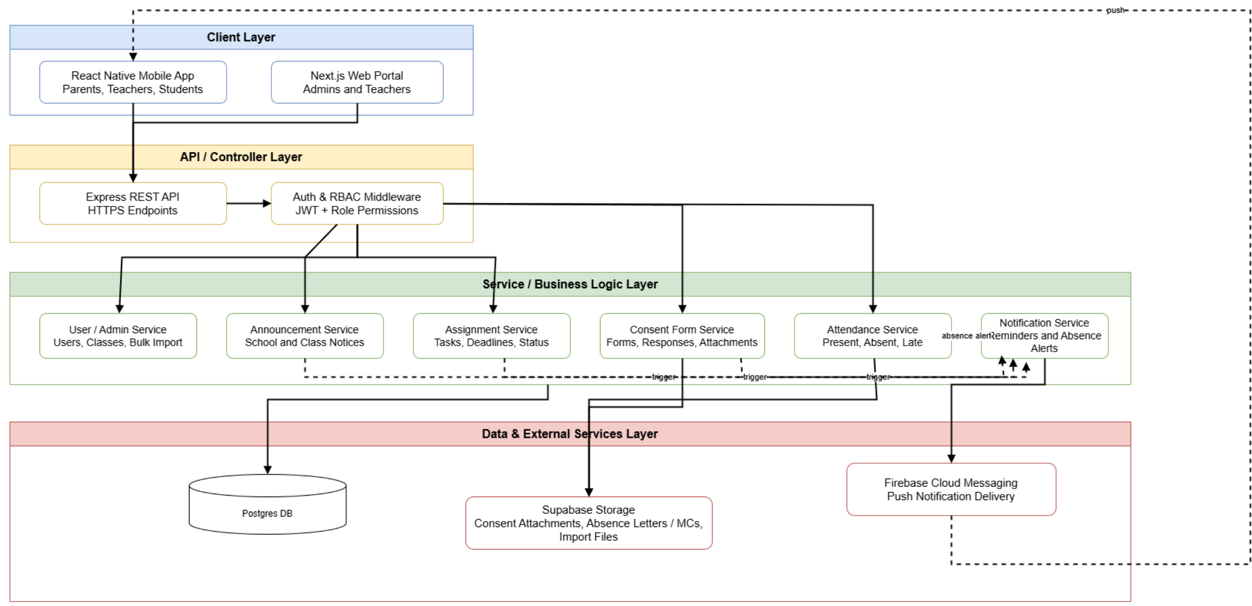
Stores:

- School policies
- Academic guidelines
- Frequently asked questions
- Administrative documents
- School announcements

This knowledge base supports the AI Assistant by providing relevant information for generating responses.

10.6 Architecture Diagram

https://drive.google.com/file/d/1rRTvjt7ISJxomNqenc1ERUP4qzC4Hw6E/view?usp=sharing



10.7 Deployment Architecture

VidyaConnect uses a cloud-based deployment architecture to ensure availability, scalability, and ease of maintenance.

11. Data Requirements

11.1 Entity Relationship Overview

VidyaConnect's data model is built around five core domains that reflect the real-world structure of a Sri Lankan school:

Identity and Access Domain
Manages all user accounts and their roles within the system — Super Admins, School Admins, Teachers, Parents, and Students.

Academic Domain
Manages the academic activities of the school — classes, assignments, submissions, marks, attendance, and the school calendar.

Communication Domain
Manages all communication between users — announcements, class messages, private messages, community groups, and notifications.

Administrative Domain
Manages school-level administrative operations — consent forms, form responses, bulk uploads, and audit logs.

Progress and Records Domain
Manages student academic progress over time — marks, term reports, achievements, and timetables.

11.2 Core Entities and Attributes

High Level Entity Relationship Diagram

The full EER diagram is available at:

https://drive.google.com/file/d/1vodNJgbJoAKkYa8PFBqIJ6wUSZQq4o4X/view?usp=sharing

11.3 Data Retention Policy

VidyaConnect stores data that is sensitive, legally significant, or operationally important. The following retention rules define how long each category of data is kept.

Data Category	Minimum Retention Period	Reason	Who Can Request Deletion
Student academic records such as assignments, marks, submissions	7 years after student leaves school	School record-keeping requirements	Super Admin only documented process

Attendance records	7 years after student leaves school	Legal and safeguarding requirements	Super Admin only
Consent form responses	3 years after form submission	Legal and administrative accountability	Super Admin only
Private messages among teacher to parent	2 years from message date	Safeguarding and audit purposes	Super Admin only(not deletable by sender)
School announcements	2 years from publication date	Communication record	School Admin can archive; Super Admin deletes
Audit logs	5 years from event date	Security and safeguarding investigation	Super Admin only immutable, no deletion
Notification history	90 days from delivery date	User reference and support	Automatically purged after 90 days
Bulk upload files	30 days after upload	Temporary processing record	Automatically purged after 30 days
User account data	Until account is deactivated + 2 years	Reactivation possibility and audit trail	Super Admin only
Password reset tokens	15 minutes from generation	Security (single use expiry)	Automatically purged on use or expiry

11.4 Data Privacy and Protection

VidyaConnect handles sensitive personal data relating to minors, families, and school staff. The following measures are applied throughout the system to protect this data.

11.4.1 Data Classification

All data stored in VidyaConnect is classified into three sensitivity levels:

Level	Category	Examples	Protection
High Sensitivity	Personal data of minors	Student names, DOB, attendance, marks, health records	Encrypted at rest, strict RBAC, audit logged
High Sensitivity	Family contact data	Parent email, phone number, home address	Encrypted at rest, visible to School Admin and linked teacher only
Medium Sensitivity	Communication data	Announcements, messages, form responses	Role-scoped access, archived with retention rules
Low Sensitivity	Operational data	School name, class names, event titles	Standard

11.4.2 Encryption

Data State	Encryption Standard	Applied To
Data in transit	All client-server communication over HTTPS	All API requests and responses
Data at rest (database)	AES-256 encryption	All PII fields in PostgreSQL
Data at rest(files)	Supabase Storage encryption	All uploaded files
File access	Signed time-limited URLs	All file downloads (no permanent public URLs)
Passwords	bcrypt with cost factor 12	All user passwords (never stored in plain text)

11.4.3 Role-Based Data Access

Access to data is enforced at both the API layer and the database query layer:

Data Type	Super Admin	School Admin	Teacher	Parent	Student
Student personal details	All schools	Own school only	Own class only	Own child only	Own record only
Attendance records	All schools	Own school only	Own class only	Own child only	Own record only
Assignment marks	All schools	Own school only	Own class only	Own child only	Own record only
Health records	Emergency access only	Designated access only	No access	Own child only	No access
Private messages	All schools — audit purpose	Own school — audit purpose	Own threads only	Own threads only	No access
Consent form responses	All schools	Own school only	No access	Own submissions only	No access

Audit logs	Full access	Own school only	No access	No access	No access

12. User Journeys

12.1 Journey 1: Parent views child's updates

Actor: Parent/Guardian
Trigger: Parent opens the app after receiving a push notification
Steps:

- Parent logs in and lands on the Parent Dashboard
- Dashboard shows unread messages, pending consent forms, upcoming assignments, and next events for the selected child
- If parent has multiple children, they use the child switcher to toggle views
- Parent taps into the announcement or assignment feed for more detail
- System marks the item as "read" and records the read timestamp
  Outcome: Parent has a consolidated, up-to-date view of their child's school activity without checking multiple channels

12.2 Journey 2: Teacher sends a class announcement

Actor: Teacher
Trigger: Teacher needs to notify all parents in an assigned class
Steps:

- Teacher logs into the web portal and opens their assigned class
- Teacher selects "New Class Announcement," writes the message, and sets a priority (normal/urgent)
- Teacher submits the announcement
- System pushes a notification to all parents linked to that class
- Teacher views the delivery/read status per parent from the class dashboard
  Outcome: All parents in the class receive the same message at the same time, with confirmation tracking

12.3 Journey 3: Teacher creates an assignment

Actor: Teacher
Trigger: Teacher needs to assign homework
Steps:

- Teacher opens the Assignment module for their class
- Teacher enters title, subject, description, deadline, and optional attachment
- Teacher submits; the assignment appears in the feed for every student/parent in that class
- System schedules an automated reminder to parents 24 hours before the deadline
- Teacher monitors submission status (Submitted / Late / Missing) from the class dashboard
  Outcome: Homework is visible to all relevant parents/students with no manual chasing by the teacher

12.4 Journey 4: Parent responds to a consent form

Actor: Parent/Guardian
Trigger: School admin distributes a new consent form (e.g., field trip permission)
Steps:

- Parent receives a notification that a new form is pending
- Parent opens the form inside the app, reviews the fields (Yes/No, text, date, file upload)
- Parent fills in required fields and submits with a digital acknowledgement/signature
- System timestamps the submission and updates the admin's live response summary
- If parent doesn't respond, system sends automatic reminders 3 days and 1 day before deadline
  Outcome: Form is completed digitally with no paper handling, and admin has real-time visibility of response status

12.5 Journey 5: Admin creates a class and links users

Actor: School Admin
Trigger: New academic year/term setup, or onboarding a new class
Steps:

- Admin logs into the web portal and opens the Class Management screen
- Admin creates a new class (name, grade, academic year) or bulk-imports class/student/parent data via Excel template
- System validates the uploaded data and flags rows with errors for correction
- Admin assigns a teacher to the class and links students to their respective parent/guardian accounts

- System auto-generates login credentials for newly created student and parent accounts
  Outcome: Class structure and role-based access are established, and all relevant users can log in and see only their own scoped data

12.6 Journey 6: Admin creates and distributes a digital consent form

Actor: School Admin
Trigger: School needs signed permission for an event (field trip, medical treatment, photography, sports day)
Steps:

- Admin opens the Consent Form module and builds a new form by selecting field types: Yes/No agreement, text response, date, or file upload
- Admin sets a submission deadline and selects the audience (by class, grade, or school-wide)
- Admin sends the form; parents receive a notification
- Admin dashboard shows a live count of submitted, pending, and overdue responses as parents respond
- Once the deadline passes, all responses lock into an archive
- Admin exports the completed response set to Excel for school records
  Outcome: The entire paper consent process is replaced with a trackable digital workflow, and admin has no manual chasing to do

12.7 Journey 7: Admin posts a school-wide announcement

Actor: School Admin
Trigger: Admin needs to communicate a school-level notice (term dates, fee reminder, sports day confirmation)
Steps:

- Admin opens the Communication Hub and selects "New School-Wide Announcement"
- Admin writes the message and sets a priority level (normal or urgent)
- Admin submits; the system distributes it to all parents, teachers, and students school-wide
- Admin views the announcement reach (read/unread breakdown) from the Admin Dashboard
- Announcement is stored in the searchable message archive
  Outcome: One authoritative message reaches every stakeholder simultaneously, replacing inconsistent teacher-relayed circulars

12.8 Journey 8: Parent sends a message to their child's teacher

Actor: Parent/Guardian
Trigger: Parent has a question or concern about their child
Steps:

- Parent opens the Communication Hub and selects the relevant teacher for their child
- Parent composes and sends a one-to-one message (not a group chat)
- System threads the message under that parent-teacher-student conversation
- Teacher receives a notification and responds
- Parent sees delivery and read status on their sent message
  Outcome: Parent gets a private, controlled channel to the teacher without cluttering a group chat or losing the conversation history

12.9 Journey 9: Student views assignments and deadlines

Actor: Student
Trigger: Student wants to check what homework is due
Steps:

- Student logs in with their own (read-only) credentials, separate from the parent login
- Student views their personal assignment feed across all subjects, filtered to Pending/Submitted status
- Student views upcoming deadlines on their personal calendar view
- Student views school and class announcements relevant to their class
- Student cannot post, message, or modify any data — access is strictly read-only
  Outcome: Student has independent visibility into their own academic obligations without depending on memory, peers, or parents relaying information

12.10 Journey 10: Parent receives and manages smart notifications

Actor: Parent/Guardian
Trigger: A scheduled digest, a risk condition, or an approaching event/deadline
Steps:

- Every Sunday evening, parent receives a weekly digest summarizing upcoming assignments and events for their child(ren)
- If a child has an overdue assignment, system sends a targeted risk alert
- System sends event reminders 3 days and 1 day before scheduled events
- Parent can open the notification log at any time to review anything missed

- Parent adjusts notification preferences per child (e.g., mute routine alerts, keep urgent ones)
  Outcome: Parent receives only relevant, well-timed notifications instead of being overwhelmed by constant WhatsApp-style noise

12.11 Journey 11: Teacher/Admin adds an event to the school calendar

Actor: Teacher or School Admin
Trigger: A new test, project deadline, sports day, exam, or meeting needs to be scheduled
Steps:

- Teacher adds a class-specific event (test, project deadline) to the calendar; Admin adds a school-wide event (term dates, sports day, exams, meetings)
- System reflects the new event immediately in the relevant calendars
- Parents/students see the event appear in their unified calendar view
- System schedules automatic reminders 3 days and 1 day before the event
  Outcome: All stakeholders see a single, consistent event schedule with no scattered or missed date announcements

12.12 Journey 12: Parent views and exports the unified calendar

Actor: Parent/Guardian
Trigger: Parent wants to plan around upcoming school and class events
Steps:

- Parent opens the Calendar module, which combines school-wide events and their child's class-specific events into one view
- Upcoming events are also highlighted on the Parent Dashboard home screen
- Parent taps "Export" to sync events to their personal Google Calendar or Apple Calendar
- If parent has multiple children in different classes/schools, calendar merges events across all of them
  Outcome: Parent has one merged calendar instead of tracking separate school notices, and can carry it into their personal device calendar

12.13 Journey 13: Each role views their Insights Dashboard

Actor: Parent, Teacher, or School Admin
Trigger: User logs in and wants a quick overview of what needs attention
Steps:

- Parent dashboard shows upcoming assignments, unread messages, pending consent forms, and next events
- Teacher dashboard shows class assignments submitted vs. outstanding, unread parent messages, and upcoming class events
- Admin dashboard shows school-wide announcement reach and outstanding consent forms
- User drills into any summary card to reach the full underlying module
  Outcome: Each role gets an at-a-glance, role-appropriate summary without navigating through every module manually

12.14 Journey 14: Admin performs bulk data upload and export

Actor: School Admin
Trigger: Start of academic year, new student intake, or exam results processing
Steps:

- Admin downloads the standard Excel template for the relevant data type (students, parents, classes, exam marks)
- Admin fills in the template and uploads it via Bulk Data Management
- System validates the upload and flags any rows with errors, showing clear messages for correction
- Admin re-uploads corrected rows; system bulk-creates parent, student, and staff logins from the validated data
- Admin later exports any module's data (students, assignments, consent responses) for school records
  Outcome: Admin sets up or maintains large volumes of school data in minutes instead of creating accounts and records one by one

12.15 Journey 15: Parent or admin experiences an offline/ low-connectivity scenario

Actor: Parent/Guardian or School Admin
Trigger: User loses mobile data connectivity mid-task (e.g., submitting a consent form)
Steps:

- User's connection drops while filling in a form or sending a message

- App queues the action locally instead of failing or losing the input
- User continues using cached data (already-loaded assignments, announcements, calendar) while offline
- Once connectivity is restored, the app automatically syncs queued actions to the server
- User is not required to repeat the action or notice that a sync occurred
  Outcome: The platform remains usable under Sri Lanka's variable connectivity conditions without data loss or duplicated effort

13. Business Rules

This section defines the core business rules that govern the behaviour of the VidyaConnect platform across all modules. Unlike functional requirements, which describe what the system does, business rules describe the policies and constraints that must always hold true, regardless of which feature or screen is being used. These rules are derived directly from the functional requirements defined in Section 6 and the role permissions defined in Section 5.

13.1 Access Control Rules

- The system shall enforce role-based access control (RBAC) at both the API layer and the UI layer, no module shall rely on UI-layer restriction alone.
- A user shall only be able to access data and features explicitly permitted to their assigned role, as defined in the Role Permission Matrix (Section 5.7).
- A Teacher shall only be able to access data belonging to their assigned class(es), access to any other class shall be denied.
- A Parent/Guardian shall only be able to access data belonging to their own linked child or children, access to any other student's data shall be denied.
- A Student shall only be able to access their own academic and attendance records, in read only mode, no student shall be able to create, edit, or delete any data.
- A School Administrator shall not be able to access or modify data belonging to another school.
- Any attempt to access data or perform an action outside a user's permitted role or school scope shall be rejected with a 403 Forbidden response and logged as a security event.
- All database queries shall include mandatory school scope filtering to prevent cross tenant data access.

- The Super Administrator shall be the only role permitted to configure system wide settings, manage School Administrator accounts, and access cross school analytics.
- Access to the AI Assistant (RAG) shall be granted to all roles, however, the data retrieved and presented by the AI Assistant for any given user shall always be limited to the data that user is independently authorised to access under their role's permissions. The AI Assistant shall never bypass or override existing RBAC restrictions.

13.2 Communication Rules

- A Teacher shall only be permitted to send messages to parents and students of their own assigned class(es) messaging parents of other classes shall not be permitted.
- A Student shall not be permitted to initiate a direct message to a Teacher or Parent.
- Messages sent outside a Teacher's defined availability window shall be queued and an automatic reply containing the Teacher's availability information shall be sent, queued messages shall be delivered at the start of the Teacher's next availability window.
- Emergency messages issued by an Administrator shall bypass all Teacher availability restrictions and shall be delivered immediately.
- Parents shall not be permitted to disable read receipts on messages sent by Administrators or Teachers.
- All messages, including private one to one conversations, shall be retained in a searchable archive for a minimum of two years and shall remain accessible to School Administrators for safeguarding purposes, even if deleted by the sender.
- Community groups shall only be created by Administrators and shall be moderated by an Administrator or a designated group moderator.
- High priority announcements marked as requiring confirmation shall remain flagged as "Action Required" for the recipient until the recipient explicitly confirms having read them.

13.3 Assignment Rules

- Only Teachers shall be permitted to create, edit, or cancel assignments, Parents and Students shall have read only access to assignment data.

- Publishing an assignment shall automatically notify all students and parents of the assigned class(es), cancelling an assignment shall trigger a cancellation notification to the same recipients.
- An assignment's submission status shall always be one of the following four defined states, Not Submitted, Submitted (on-time), Late, or Missing.
- A Student shall not be permitted to resubmit an assignment after its deadline has passed, unless the assigned Teacher explicitly re enables submission for that assignment.
- A Teacher shall retain the authority to manually override a system assigned submission status where appropriate.
- An automated reminder shall be sent 24 hours and 1 hour before an assignment deadline if the assignment has not yet been submitted, unless the Teacher has explicitly disabled reminders for that assignment.

13.4 Attendance Rules

- Only Teachers shall be permitted to mark attendance for their assigned class(es), once submitted, an attendance record shall be locked and shall only be amended following an explicit request to, and approval from, a School Administrator.
- Attendance status shall always be recorded as one of the following defined values: Present, Absent, Absent–Excused, Late, or Early Departure.
- Where a student is marked "Absent," an automatic notification shall be sent to the linked parents within 5 minutes of the record being submitted.
- A parent shall be permitted to submit a justification for an absence within 3 calendar days of the absence date, submissions after this period shall not be accepted through the standard justification workflow.
- An absence justification shall remain in "Pending Review" status until a Teacher either accepts it (updating the record to "Absent Excused") or rejects it with a stated reason, the parent shall be notified of the outcome in either case.
- Where a student has 3 or more consecutive absences due to illness, the parent shall be prompted to upload a supporting medical certificate, access to uploaded medical certificates shall be restricted to the assigned Teacher and School Administrator only.
- A low attendance alert shall be automatically triggered whenever a student's monthly attendance falls below 80%, and shall be sent to the parent, the assigned Teacher, and the School Administrator; the alert shall be escalated if attendance remains below 80% for two consecutive months.

13.5 Consent Form Rules

- Only School Administrators shall be permitted to create and distribute digital consent and admin forms.
- A consent form shall not be distributed to any account that is marked inactive.
- Once a parent submits a response to a consent form, the response shall not be editable unless the School Administrator explicitly re opens the form for that parent.
- Formal acknowledgement forms (such as policy or fee acknowledgements) shall require an explicit statement and checkbox confirmation, and the submission timestamp and parent identity shall be permanently recorded for audit purposes.
- Automated reminders for outstanding consent forms shall be sent at 3 days, 1 day, and on the day of the submission deadline, unless the School Administrator has explicitly disabled automated reminders for that specific form.
- All consent form responses shall be retained in a searchable archive for a minimum of three years.
- Fee payment acknowledgement forms shall be categorised separately from general consent forms within the archive, and any outstanding fee acknowledgements shall be flagged on the Administrator dashboard.

13.6 Notification Rules

- All push notifications shall be delivered within 60 seconds of the triggering event under normal operating conditions; emergency alerts shall be delivered within 30 seconds.
- Emergency alerts shall bypass all individual user notification preferences and shall always be delivered, users shall not be permitted to disable emergency alerts.
- The Weekly Digest notification shall be sent every Sunday at 6:00 PM and shall summarise upcoming assignments, events, unread announcements, and pending forms, a parent may opt out of the Weekly Digest while retaining all other notification types.
- A Risk Alert shall be automatically triggered when any of the following conditions are met for a student: two or more overdue assignments, attendance below 80%, or an unread announcement older than 48 hours, Risk Alerts shall require explicit acknowledgement from the relevant Teacher or Administrator.
- Where push notification delivery via Firebase Cloud Messaging is unavailable, the system shall fall back to logging the notification in the user's in app notification log.
- Notification history shall be retained and accessible to each user for a minimum of 90 days.

- School Administrators shall have the authority to designate certain notification types as mandatory school wide, such notification types shall not be configurable or disabled by individual users.

13.7 Calendar and Conflict Rules

- Only School Administrators shall be permitted to create school wide calendar events, Teachers shall only be permitted to create events scoped to their own assigned class(es).
- Where a new event is scheduled on a date or time that conflicts with an existing event, the system shall display a warning to the creating user, the system shall not block creation of the conflicting event.
- Where an event or assignment is scheduled on a date defined as a public holiday or school closure, the system shall display a warning to the creating user, the system shall not block creation.
- Only School Administrators shall be permitted to publish or modify the official exam timetable, Teachers shall not have edit access to exam timetable entries.
- Where RSVP is enabled for an event, a parent shall be permitted to change their RSVP response up to 24 hours before the scheduled event start time, changes after this window shall not be accepted.
- Calendar export to external personal calendars (Google Calendar, Apple Calendar) shall use the standard iCalendar (.ics) format to ensure compatibility.

13.8 Data Management Rules

- Bulk data uploads shall be validated in full prior to import, if any row in the uploaded file contains an error, the entire batch shall be rejected and no partial commit of valid rows shall occur.
- A detailed error report shall be generated for any failed bulk upload, specifying the row number, affected column, and reason for each validation failure.
- Bulk imports of parent and staff data shall automatically attempt to link parent accounts to the corresponding student records using the Student ID field, duplicate email addresses shall be flagged rather than silently overwritten.

- All bulk data exports (student lists, attendance records, consent form responses, exam marks, etc.) shall include a cover sheet specifying the school name, export date, applicable date range, and total record count.
- Large data exports shall be processed asynchronously, and the requesting Administrator shall be notified upon completion rather than the export blocking the user interface.
- Data retained for audit, safeguarding, or legal purposes (message archives, consent form responses, attendance records, audit logs) shall not be permanently deleted by any user role, including School Administrators, deletion of such records, where legally required, shall only be performed by the Super Administrator through a documented process.

13.9 Account Management Rules

- User accounts shall not be self registered by parents, teachers, or students, all accounts shall be created by a School Administrator or Super Administrator, either individually or via bulk upload.
- All first time users shall be required to change their system generated temporary password upon initial login before accessing any other part of the system.
- An account shall be automatically locked for 15 minutes following 5 consecutive failed login attempts.
- A password shall not be reused if it matches any of the user's previous 5 passwords.
- A parent account shall be permitted to link to multiple children however, each student record shall be linked to a maximum of 2 parent/guardian accounts.
- Linking a new parent account to an existing student record shall require explicit approval from a School Administrator before the link becomes active.
- Where a user's role is changed, all active sessions and refresh tokens associated with that user shall be immediately invalidated, requiring re authentication under the new role.
- A School Administrator shall have the authority to force a password reset for any user within their own school, this authority shall not extend to users of other schools.
- Student account access may be disabled school wide at the discretion of the School Administrator where disabled, all student credentials shall remain separate from and unaffected by parent account credentials.

14.Assumptions and Constraints

14.1 Assumptions

Devices & Hardware

- All teachers and school administrators have access to at least one Android smartphone or a desktop/laptop with a modern browser (Chrome, Firefox, Edge) to access the web portal.
- Parents and guardians own or have regular access to an Android or iOS smartphone capable of running React Native applications.
- Students are assumed to be secondary-level; their app access is read-only (view announcements, assignments, notifications).
- The school has at least one designated computer or tablet available for admin-level bulk data operations.

Internet Connectivity

- Schools have at least intermittent internet connectivity (2G/3G minimum) sufficient for sending and receiving push notifications and small data payloads.
- The system does not assume continuous high-speed internet. However, full offline-first sync is deferred post-MVP. The app will cache recently loaded data for read access when offline, but write operations (marking attendance, submitting forms) require an active connection.
- Teachers are assumed to have mobile data or school Wi-Fi available during school hours when marking attendance.

School Administration

- School has at least one designated School Administrator who is responsible for setting up the system — creating classes, importing users, and managing roles.
- Student and parent data (names, contact numbers, class assignments) already exists in some form (paper records or a spreadsheet) and can be structured into the provided Excel bulk import template.
- Each student is linked to at least one parent or guardian account in the system.
- Each class has exactly one assigned class teacher at any given time.
- The school follows a standard academic year structure with defined classes and grade levels.

User Access & Accounts

- Every user (admin, teacher, parent, student) has a unique account created by the school administrator, either manually or via bulk import.
- Login credentials are distributed to users by the school administrator through existing school communication channels (printed slips, direct handover) during onboarding.
- Users are assumed to have basic smartphone literacy — ability to install an app, log in, read notifications, and tap buttons. No advanced technical knowledge is assumed.
- A parent or guardian may be linked to more than one student (siblings in the same school).
- A teacher may be assigned to multiple classes but is the class teacher of only one class.

Data

- School administrators are responsible for the accuracy of data entered during initial setup (student names, class assignments, parent contact details).
- The system does not integrate with any existing school management system (SMS) or government education database in the MVP. All data is entered manually or via bulk import.

14.2 Constraints

Technical Constraints

- The mobile application is built using React Native and must run on Android 8.0+ and iOS 13+. Older devices are not supported.
- File uploads (absence letters, MC documents, consent attachments) are limited to a maximum of 5 MB per file to stay within Supabase Storage free tier limits during the pilot.
- The bulk import Excel template supports a maximum of 1,000 rows per upload in the MVP to prevent server timeout during processing.
- Full offline-first bi-directional sync (WatermelonDB) is out of scope for the MVP. Only basic read caching and queued write operations are implemented.

Security & Privacy Constraints

- All data transmitted between the client and server must be encrypted using HTTPS/TLS. No plain HTTP is permitted.
- JWT tokens expire after 24 hours; users must re-authenticate after expiry.
- Absence letters and medical certificates uploaded by parents are stored securely in Supabase Storage with signed URL access; they are not publicly accessible.

- The system must not expose student personal data (full name, class, health details) to users of a different school even if hosted on the same backend instance.

Regulatory & Institutional Constraints

- The system operates within Sri Lankan school structures and must respect the Ministry of Education's guidelines on student data handling where applicable.
- The pilot must be conducted with parental consent for storing student data, coordinated by the school administration.

Project Constraints

- The technology stack is fixed as React Native, Next.js, Node.js/Express, and PostgreSQL.
- Third-party integrations beyond FCM and Supabase Storage are out of scope for the MVP (no payment gateway, no Google Calendar sync, no LMS integration).
- The MVP architecture must support a single pilot school of up to 800 students without requiring re-architecture for future multi-school scalability, as defined in Section 7.5 (Scalability Requirements); however, full multi-tenant validation at scale (≥50 schools) is out of scope for the current phase.

15.Risks

This section identifies the key risks that could affect the successful delivery of VidyaConnect, along with the corresponding mitigation strategies. Each risk is rated by likelihood and impact to support prioritisation, and is paired with a concrete mitigation approach rather than a general statement of awareness.

15.1 Technical Risks

	Risk	**Likelihood**	**Impact**	**Mitigation**
1	The mobile application performs poorly on low-end Android devices (2GB RAM), resulting in slow load times or crashes for the majority of	Medium	High	Conduct performance testing early and continuously on low-spec Android devices rather than only flagship devices; avoid heavy

	parent users.			libraries; enforce the data-usage and performance budgets defined in Section 7.1.
2	Offline-first synchronisation fails to reconcile correctly when a device reconnects after an extended offline period, resulting in data loss or duplicate records (e.g. duplicate attendance entries, lost form submissions).	Medium	High	Implement and test a clear conflict-resolution strategy (e.g. last-write-wins with timestamps, or queued-action replay) before any offline-capable feature is marked complete; include offline/reconnect scenarios explicitly in the test plan.
3	Push notification delivery fails or is delayed due to Firebase Cloud Messaging outages, network instability, or platform-level restrictions (e.g. Android battery optimisation killing background processes).	Medium	Medium	Implement the in-app notification log as a guaranteed fallback (Section 6.4.7) so no notification is fully lost even if push delivery fails; monitor delivery success rate and alert if it drops below the 95% target (Section 7.1.4).
4	The chosen file storage provider (Supabase Storage vs. AWS S3) is not finalised, creating a risk of rework if the decision is made late in development.	Medium	Medium	Finalise the file storage decision with mentor input early in Semester 1, before any file-upload-dependent feature (consent forms, assignment submissions, health records) is built against a specific provider's API.
5	Database schema changes required mid-project (e.g. to support a feature not fully scoped at design time) cause significant rework across both backend and already-completed frontend UI.	Medium	Medium	Finalise the core database schema (ERD) before Semester 1 frontend UI work begins for each module; treat schema changes after this point as requiring formal review before implementation.
6	Third-party service free-tier limits (Firebase, hosting platform, file storage) are exceeded during pilot testing, causing service disruption.	Low	Medium	Monitor usage against free-tier quotas throughout pilot testing; identify the upgrade path and cost in advance so the team is not blocked if a quota is reached.

15.2 Adoption Risks

	Risk	Likelihood	Impact	Mitigation
1	Parents with low digital literacy struggle to use the application, leading to low engagement and a return to informal channels (WhatsApp, paper) alongside or instead of VidyaConnect.	High	High	Apply the icon-driven, minimal-text UI approach defined in Section 7.4.4 from the outset; conduct usability testing with representative parent users (not just the development team) before pilot deployment; keep onboarding to a maximum of 4 screens (Section 7.4.3).
2	Teachers perceive the platform as an additional administrative burden on top of existing manual processes, rather than a replacement for them, leading to inconsistent usage.	Medium	High	Ensure the platform clearly replaces (not duplicates) existing manual tasks — e.g. once attendance is digital, paper attendance registers should be discontinued at the pilot school; involve teachers in usability feedback sessions before full rollout.
3	The pilot school's existing communication habits (WhatsApp groups, printed circulars) persist in parallel with VidyaConnect, fragmenting communication further rather than consolidating it.	Medium	High	Secure explicit commitment from the pilot school's administration to designate VidyaConnect as the official communication channel during the pilot period, with clear communication to parents and staff about the transition.
4	Multi-child families find switching between children's profiles confusing or cumbersome, reducing the platform's value for the parents who need it most.	Low	Medium	Validate the multi-child switching flow (Section 6.1.5) specifically with multi-child parent users during usability testing, rather than assuming single-child usage patterns are representative.
5	Low initial account activation rates after bulk account creation, if parents do not receive or act on their login credentials.	Medium	Medium	Provide School Administrators with a clear, printable credential distribution process; consider a guided first-login flow with in-person support during the pilot school's initial rollout.

15.3 Data Privacy Risks

	Risk	Likelihood	Impact	Mitigation
1	Sensitive student data (academic records, attendance, health information, family contacts) is exposed due to a role-based access control flaw, allowing a user to view data outside their authorised scope.	Low	Critical	Enforce RBAC at both the API and UI layers as a mandatory architectural rule (Section 13.1); include explicit cross-role and cross-school access tests in the test plan; log and alert on any 403 Forbidden events as potential security incidents (Section 7.2.3).
2	Absence of a fully enforced Sri Lankan data protection law results in inconsistent or unclear data handling practices, since there is no binding local legal standard to follow.	Medium	Medium	Continue to apply GDPR-aligned principles as the project's internal data privacy baseline (Section 14.2); monitor for any future Sri Lankan data protection legislation that may require the platform to adapt its data handling practices.
3	The AI Assistant (RAG-based) retrieves or surfaces data the requesting user is not authorised to see, due to a flaw in the retrieval pipeline's scoping logic.	Low	Critical	Enforce role-based filtering at the data retrieval stage of the RAG pipeline itself, not only at the final response stage; treat this as a mandatory architectural constraint (Section 13.1) and test explicitly for cross-student and cross-role data leakage before any AI Assistant functionality is considered complete.
4	Uploaded files (consent form attachments, medical certificates, assignment submissions) are stored without adequate access control, exposing sensitive documents via guessable or unsecured URLs.	Low	High	Use signed, time-limited URLs for all file access (Section 7.2.1), regardless of which storage provider is ultimately selected (Supabase Storage or AWS S3); never expose permanent public file links.
5	Audit logs themselves are tampered with or deleted, undermining the platform's ability to investigate security or safeguarding incidents after the fact.	Low	High	Implement audit logs as immutable records with no deletion or editing capability for any role, including School Administrators (Section 7.2.4, Section 13.8); restrict audit log access to Super Administrator

				and School Administrator roles only, as already defined.

15.4 Scope Risks

	Risk	Likelihood	Impact	Mitigation
1	Feature scope expands beyond what is achievable within the one-year academic timeline ("scope creep"), particularly as new feature ideas are suggested during development (e.g. the addition of the AI Assistant after the original discovery phase).	High	High	Maintain strict adherence to the Feature Scope Classification (Section 8); any new feature request must be formally evaluated and assigned to Core MVP, Extended Scope, or Optional/Future Scope before any development time is allocated to it.
2	Frontend UI is completed in Semester 1 for features whose requirements later change during Semester 2 backend development, causing UI/backend mismatch and rework (see Section 14.3, UI/backend drift risk).	Medium	Medium	Freeze functional requirements for each feature (Section 6) before its UI is built in Semester 1; treat any requirement change after UI completion as a formal change request requiring impact assessment on already-completed work.
3	The AI Assistant's UI is built in Semester 1 ahead of its backend, creating pressure to complete its full RAG-based functionality within the project timeline despite its Optional/Future Scope classification (Section 8.3).	Medium	Medium	Clearly communicate to all stakeholders (mentors, pilot school, team) that AI Assistant UI completion does not equate to functional completion; treat its backend as genuinely optional, dependent on time remaining after Core MVP and Extended Scope are fully delivered.
4	Extended Scope features (Section 8.2) are not fully backend-integrated by the end of Semester 2, leaving some UI screens non-functional at project completion.	Medium	High	Prioritise Extended Scope features within Semester 2 according to the order listed in Section 8.2 (which reflects justification/value), so that if time runs short, the lowest-priority Extended Scope features are the

				ones left incomplete, not core ones.
5	Pilot school feedback during Semester 2 surfaces new requirements that do not fit cleanly into the existing scope classification, creating pressure to accommodate late-stage changes.	Medium	Medium	Log all pilot feedback as candidate items for future versions (post-project) by default; only incorporate Semester 2 feedback into current scope if it represents a critical defect or safety issue, not a feature enhancement.

15.5 Team Risks

	Risk	**Likelihood**	**Impact**	**Mitigation**
1	The team lacks a dedicated DevOps or infrastructure specialist, increasing the risk of deployment, environment configuration, or CI/CD issues consuming disproportionate development time.	Medium	Medium	Rely on managed, low-configuration services (cloud PaaS hosting, Firebase, GitHub Actions) as already reflected in the technology stack (Section 6, System Architecture) specifically to minimise the DevOps burden on the team.
2	Uneven distribution of frontend vs. backend skills across team members results in one area (e.g. backend APIs) lagging significantly behind the other (e.g. UI), creating the UI/backend drift risk described in Sections 14.3 and 15.4.	Medium	Medium	Assign clear ownership of frontend and backend workstreams early, with cross-training where possible; track Semester 1 progress against both the UI completion goal and the Core MVP backend goal separately, not as a single combined metric.
3	Key team member unavailability (illness, academic conflicts with other modules, personal circumstances) during a critical development period delays delivery.	Medium	Medium	Maintain shared documentation (Section 7.6.1) so any team member can pick up another's work if needed; avoid single points of ownership over any one module where feasible.
4	Reliance on mentor feedback for open decisions (file storage	Medium	Medium	Raise open decision items with mentors as early as possible in

	provider, LLM provider, revenue model) introduces schedule dependencies outside the team's direct control, as already noted in Section 14.4.			Semester 1 rather than waiting until the dependent feature is scheduled for development; default to a reasonable placeholder decision if mentor feedback is delayed, to avoid blocking progress.
5	Team members' unfamiliarity with newer or less-used parts of the stack (e.g. RAG architecture and vector databases for the AI Assistant) leads to underestimated development time if and when that work begins.	Medium	Low	Treat the AI Assistant's technical approach as requiring dedicated research and learning time before implementation begins, separate from and in addition to its feature development time, consistent with its Optional/Future Scope status (Section 8.3).

16. MVP Acceptance Criteria

The MVP will be considered complete when the following conditions are met, organized by module:

16.1 Communication Hub

- School-wide, class-level, and one-to-one messages can be created, sent, and received by the correct roles only
- Read receipts are recorded and visible to the sender
- All sent messages are stored and searchable by date and sender

16.2 Assignment and Task Tracking

- Teachers can create assignments with title, subject, deadline, and description
- Parents and students see only assignments relevant to their own child/class
- Submission status (Submitted/Late/Missing) updates correctly
- Automated 24-hour-before-deadline reminders are delivered

16.3 Smart Notifications

- Weekly digest is generated and sent every Sunday evening
- Risk alerts trigger correctly when a student has an overdue assignment

- Notification log is accessible and accurately reflects past notifications

16.4 Digital Consent and Admin Forms

- Admin can create a form with at least the four supported field types (Yes/No, text, date, file upload)
- Parents can submit a form and receive confirmation
- Admin dashboard shows accurate live counts of submitted/pending/overdue responses
- Reminder notifications fire at 3 days and 1 day before deadline
- Responses can be exported to Excel after the deadline

16.5 Calendar and Events

- Admin and teacher can post school-wide and class-specific events respectively
- Parent calendar correctly merges school and class events for their child
- Calendar export to Google/Apple Calendar works without data loss

16.6 Insights Dashboard

- Each role (parent, teacher, admin) sees a dashboard populated with real, correct data for their scope only

16.7 Bulk Data Management

- Admin can bulk-upload students, parents, and staff via the provided Excel template
- Invalid rows are flagged with clear, specific error messages
- Bulk-created accounts can log in successfully with correct role permissions

16.8 Cross-cutting / Non-Functional

- Role-based access control is enforced everywhere: no role can see or modify data outside its scope
- Core features (viewing assignments, announcements, forms) remain usable when the app goes offline and sync automatically without data loss once connectivity returns
- The app runs acceptably on a 2GB RAM, Android 8+ device
- Student accounts are strictly read-only with no ability to post or modify data